FOR AGENDA

EBS/06/132

October 18, 2006

To: Members of the Executive Board

From: The Secretary

Subject: **Ghana—Sixth and Final Review Under the Three-Year Arrangement Under the Poverty Reduction and Growth Facility and Request for Waiver of Nonobservance of Performance Criterion**

Attached for consideration by the Executive Directors is a paper on the sixth and final review under the three-year arrangement under the Poverty Reduction and Growth Facility for Ghana and its request for a waiver of nonobservance of a performance criterion, which is tentatively scheduled for discussion on **Friday, October 27, 2006**. A draft decision appears on page 16. At the time of circulation of this paper to the Board, the Secretary's Department has received a communication from the authorities of Ghana indicating that they consent to the Fund's publication of this paper.

Questions may be referred to Mr. Itam (ext. 35350), Ms. Pattillo (ext. 37319), and Ms. Loukoianova (ext. 35916) in AFR.

Unless the Documents Section (ext. 36760) is otherwise notified, the document will be transmitted, in accordance with the procedures approved by the Executive Board and with the appropriate deletions, to the WTO Secretariat on Thursday, October 26, 2006; and to the African Development Bank, the European Commission, and the Organisation for Economic Cooperation and Development, following its consideration by the Executive Board.

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GHANA

Sixth and Final Review Under the Three-Year Arrangement Under the Poverty Reduction and Growth Facility and Request for Waiver of Nonobservance of Performance Criterion

Prepared by the African Department
(In consultation with other departments)

Approved by Saul Lizondo and Adrienne Cheasty

October 18, 2006

- Discussions for the sixth and final review under the Poverty Reduction and Growth Facility (PRGF) arrangement were held in Accra August 21–September 4, 2006. The staff team consisted of Mr. Itam (head), Ms. Loukoianova, and Ms. Pattillo (all AFR), Mr. Akitoby (FAD), Ms. Brixiova (PDR), and Ms. Cheng (assistant, AFR). The mission was assisted by Mr. McIntyre, resident representative in Accra. Mr. Kwakye (OED) participated in the discussions. The mission coordinated with teams from the World Bank and the donor community.
- The mission met with Minister of Finance and Economic Planning Baah-Wiredu, Bank of Ghana Governor Acquah, Deputy Finance Minister Osei, and other officials and private sector representatives. At their request, the mission paid courtesy calls on H.E. President Kufuor and H.M. the Asantehene Otumfuo Osei Tutu II.
- A three-year PRGF arrangement in support of Ghana's Poverty Reduction Strategy (GPRS) was approved in May 2003 in the amount of SDR 184.5 million (50 percent of quota). The arrangement was later extended to October 31, 2006 (EBS/05/79). The fourth and fifth reviews were completed in June 2006. Ghana reached the completion point under the enhanced Heavily Indebted Poor Countries (HIPC) Initiative in July 2004. Also, the country has received further debt relief under the Multilateral Debt Relief Initiative (MDRI) from the Fund, the IDA, and the African Development Fund.
- Ghana's Poverty Reduction Strategy (GPRS II) was updated in January 2006 (EBS/06/23 and EBD/06/24). The Board considered the Joint Staff Advisory Note (EBD/06/50) in June 2006.
- Relations with the Fund are summarized in Appendix I, and IMF-World Bank collaboration and Ghana's financial relations with the World Bank Group in Appendix II. This report is accompanied by a letter of intent from the Ghanaian authorities (Appendix IV).
- It is expected that missions for the Ex-Post Assessment and 2007 Article IV consultation will be later this year and early next year, respectively, with Executive Board consideration of the reports around April 2007.
- The authorities have agreed to publish both the staff report and the letter of intent.

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EXECUTIVE SUMMARY

Ghana is achieving the macroeconomic objectives of its poverty reduction strategy, supported by the PRGF arrangement. Growth is relatively strong, inflation is falling, and the external position has strengthened considerably. Reflecting the improvement in economic conditions, the growth of real GDP per capita has more than doubled over the last five years. Based on current projections, the poverty rate would be halved before 2015.

The policy implementation underpinning the strong economic performance is satisfactory. Fiscal and monetary performance in the first half of 2006 was in line with the program, and the near term prospects are for further macroeconomic consolidation. There has been progress with structural reforms to enhance the performance of public enterprises, raise the productivity of the civil service, deepen financial intermediation, and improve economic governance. Continued progress in improving the policy environment would strengthen the policy underpinning for strong growth in the period ahead.

Ghana's program implementation through June 2006 was satisfactory, although some earlier slippages had to be corrected over the last two years. Only one of the quantitative and structural performance criteria for June 2006 (net domestic assets of the Bank of Ghana) was not observed, and that was for technical reasons. Staff supports the authorities' request for a waiver and recommends completion of the sixth and final review under the PRGF arrangement.

Ghana has established mechanisms to track the use of the resources released by debt relief provided by the MDRI. For 2006, a supplementary budget allocates the released resources almost equally between poverty-related expenditure and growth-enhancing investment; amounts for 2007-08 will be incorporated into the annual budgets.

Based on their scaled-up investment plan and policy reforms, the authorities project an acceleration in real growth rates over the medium term. The two major risks to this scenario are that (i) the resources available may not be sufficient to finance the scaled-up investment plan, resulting in gaps in critical infrastructure and lower growth; and (ii) the increase in investment may not push the growth trajectory up as far as expected.

In the authorities' high growth scenario with some nonconcessional borrowing to finance the investment plan, Ghana has a moderate risk of debt distress. However, there is a risk that the expected growth may not be realized, thereby jeopardizing the debt sustainability, as lower growth would cause the debt indicators to worsen. The authorities stressed that to keep debt sustainable they will exercise great caution in contracting new loans, carefully monitor debt accumulation, and use all resources productively—including limited nonconcessional borrowing for high-return projects.

I. Introduction

1. **Since 2003 Ghana's economic performance has been quite satisfactory, despite some early slippages, which have been corrected**. The country is achieving the macroeconomic objectives of its poverty reduction strategy, thanks to a marked improvement in macroeconomic management and a favorable external environment. Real GDP growth has exceeded the target for the last two years. The growth of real GDP per capita—a core indicator of the authorities' program—has in turn doubled, with per capita income rising in constant 2000 US dollar terms to US$306 in 2005 from US$270 in 2000. Although the process has not always been smooth, inflation is now heading downward. The stability of the nominal exchange rate against the U.S. dollar has also helped dampen inflation. The real effective exchange rate for the cedi appreciated significantly in the last two years, mainly reflecting an inflation differential with major trading partners (Figure 1). Public debt has declined substantially as a result of prudent macroeconomic management and debt relief under the enhanced Highly Indebted Poor Countries (HIPC) Initiative and the MDRI. Structural reforms have begun to yield benefits, particularly in agriculture and services.

II. Recent Economic Developments and Immediate Prospects

A. Recent Developments

2. **Economic performance continued to be generally satisfactory in the first half of 2006.** Real GDP growth is estimated to have been around 6 percent, with strong expansion in agriculture and mining (Table 1). Twelve-month inflation declined from 14.8 percent at the end of 2005 to 10.5 percent in June 2006, even with significant upward adjustments of domestic petroleum prices[1] (Figure 2). The external current account deficit narrowed somewhat, aided by production from a new gold mine, an upswing in cocoa exports, continued strong remittance inflows,[2] and significant improvement in the terms of trade (Table 2). The terms of trade improvement in 2006 is now projected to be about 7 percent compared to the 3 percent envisaged under the program, mainly because the increases in commodity prices (particularly for cocoa and gold) have been higher, more than offsetting the increases in import prices (particularly for oil). These developments, combined with continued net capital inflows, made possible further accumulation of gross international reserves, which rose to US$2 billion at the end of June, equivalent to about 2½ months of imports of goods and services.

3. **The stability of the nominal exchange rate against the U.S. dollar has helped to keep inflation on the decline.** In 2006, the cedi-U.S. dollar exchange rate stayed largely unchanged. However, the nominal effective exchange rate for the cedi depreciated by 5 percent by the end of June 2006. This and a reduction in the inflation differential with major trading partners combined in the first half of 2006 to yield a depreciation of the cedi in

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[1] The recent pick-up in inflation was due entirely to the continuing rise in domestic petroleum product prices in line with increases in world oil prices.

[2] See Appendix III for recent trends in remittances to Ghana.

real effective terms of about 2 percent. The mission observed that the relative stability of the cedi-U.S. dollar exchange rate suggests that the de facto exchange rate regime is akin to a conventional fixed peg. The authorities argued that the stability of the cedi vis-à-vis the U.S. dollar over the last few years reflects primary the strong economic fundamentals and success in macroeconomic stabilization. Their financial program is based on a monetary anchor that allows monetary management to aim at controlling liquidity. Therefore, the authorities would describe their exchange rate regime as a managed float with no preannounced path.

4. **Fiscal performance in the first half of 2006 was in line with the program.** The deficit after grants and discrepancy was 2.6 percent of GDP, compared to the 4.1 percent of GDP the program projected (Table 3). Revenue was slightly higher than expected because collection of direct and petroleum taxes was better. Grants were also higher than expected as IMF MDRI resources were transferred from the Bank of Ghana. Meanwhile, expenditure (including discrepancy) was 0.3 percent of GDP lower than planned, mainly because capital projects were executed more slowly than anticipated. Current spending was on target; savings on wages and salaries were offset by unanticipated petroleum-related subsidies.[3] MDRI expenditure amounted to 0.2 percent of GDP in the first half of the year.

5. **At the end of June 2006, monetary aggregates were in line with the program.** Reserve money grew at the planned annualized rate of 19.7 percent; broad money growth at 28.6 percent, however, was faster than planned (Table 7). With inflation expectations favorable and with relatively low government borrowing on the domestic market, most interest rates continued to decline (Figure 3).

B. Immediate Prospects

6. **The immediate prospects are for economic activity to continue at the current pace.** Real GDP growth in 2006 should hold at around 6 percent, although there remains a short term risk associated with an energy crisis that emerged in September 2006.[4] However, recent upward adjustments of domestic petroleum product prices as costs are fully passed through are likely to make it impossible for Ghana to achieve single-digit inflation by the end of the year; inflation is now projected to be slightly over 10 percent. The external current account deficit (including official transfers) is projected to be about 5½ percent of GDP,

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[3] These were one off arrears to the oil marketing companies—for the delay in adjusting prices in the initial stages of the new mechanism—that were due in 2005 but the bills were not verified until 2006.

[4] Low water levels at the Akosombo Dam, the largest of the two hydro electric generating dams in Ghana, are the proximate cause of the recent energy crisis. Electricity demand has also increased substantially in the last decade in line with increased economic activity, a large increase in access associated with the national electrification system, and most recently, large consumption needs of VALCO, the aluminum smelter. In response to the crisis, the generating and distribution companies instituted a national load management exercise beginning at the end of August 2006. The authorities are undertaking measures to rectify the situation, which they expect will have an impact in the next few months.

1 percentage point of GDP lower than the program expected. Gross international reserves should be about the US$2 billion targeted.

7. **On the fiscal side, net domestic financing of the budget—the key indicator of fiscal performance under the program—is expected to hit the program target**. This will allow the domestic debt-to-GDP ratio to decline further to 8.3 percent in 2006 from 10.8 percent in 2005. The overall budget deficit (after grants and discrepancy) will probably be larger than envisaged because critical infrastructure projects are being executed with more concessional foreign financing than expected.

8. **On the monetary side, the central bank is expected to continue its efforts to control liquidity and contain inflation**. Projections indicate that reserve money growth (the operational target) would continue along the program path. Also, broad money expansion will revert to past trend—following a sharp deceleration with the unusual slowdown in government spending in the last quarter of 2005—and the 2006 target should be achieved.

III. Program Implementation

9. **Ghana's program implementation in 2006 through June was satisfactory.**[5] All but one of the quantitative performance criteria for June 2006 were met (Appendix IV, Table 1). The single exception was the ceiling on net domestic assets of the Bank of Ghana, which was exceeded by a relatively small margin for technical reasons. The Technical Memorandum of Understanding inadvertently omitted an adjuster for the MDRI resources that were being spent and absorbed. If that adjuster had been included as intended, the performance criterion would have been met.[6] In the circumstances, staff supports the authorities' request for a waiver of the nonobservance of the ceiling on the net domestic assets of the central bank (Appendix IV).

10. **The structural performance criterion and benchmark for June 2006 were met** (Appendix IV, Table 2). A government report on the recent comprehensive review of the compensation structure for public sector workers will form the basis for rationalizing compensation, monetizing some nonmonetary benefits, standardizing job titles, and updating job classifications. A Commission on Wages and Salaries has been set up to devise the government's new wage policy. It is expected that the new comprehensive pay structure will be announced in next year's budget, which will take effect in January 2007. Also, in line with Ghana's strategy for financial sector reform, the cabinet has approved the credit reporting bill and submitted it to parliament, which is expected to pass it by the end of the year.

11. **Structural reform is a prominent component of the government's strategy for economic development.** The authorities have focused on enhancing the operations and

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[5] See the authorities' letter of intent in Appendix IV.

[6] The performance criteria for net domestic financing of government and net international reserves were also affected by the omission of the adjuster. However, the net international reserves overperformed even after correction for this omission.

finances of public enterprises, raising the productivity of civil service, deepening financial intermediation, and improving governance generally. The financial performance of state-owned enterprises (SOEs) has improved. Also, the State Enterprise Commission (SEC) has signed contracts with 31 out of 35 SOEs to foster accountability and monitor their financial and operating performance. SOEs must now report quarterly, and there is a high rate of compliance. However, progress on setting up a comprehensive database of SOEs has been somewhat slow because of staff shortages and difficulties in retaining skilled accountants at the SEC.

12. **The new computerized payroll management system (IPPD2) is being deployed and subvented agencies are being integrated.** The Ministry of Health has been integrated into it and remaining ministries, departments, and agencies are to be phased in by the end of October 2006. Moreover, 80 percent of the wage bill of the 137 subvented agencies will be integrated into the IPPD2 by the end of this year.

13. **The Financial Sector Strategic Plan (FINSSP) guides reform in the financial sector.** An important structural reform measure in 2006 was the elimination of the secondary reserve requirement for deposit money banks in August 2006.[7] This is expected to make the banking system more efficient. Partly to help mobilize financial savings in the face of increasing competition, commercial banks started to remove minimum deposit requirements in the first half of 2006, which has had immediate impact in enhancing financial deepening. All commercial banks, except for two small ones, will meet the minimum capital requirement that comes into effect on December 31, 2006; the small banks are being encouraged to consider mergers as an option. Also, parliament is expected to pass a foreign exchange bill in October 2006 that will allow foreign investors to participate in the secondary domestic capital market. Nearly all the key recommendations of the FSAP update have been implemented.

14. **Economic aspects of governance have been improving:**

- **Fiscal reporting is more timely and accurate.** Since August 2005, monthly reports on budget execution have generally been completed within six weeks, although more work is needed to reduce discrepancies between above- and below-line data and to fully cover externally financed projects.

- **The government has made progress in implementing the computerized budget and public expenditure management system (BPEMS).** It is expected that within weeks all budgetary operations in the eight pilot MDAs will be conducted in BPEMS. In addition, a decentralized payment system is being introduced to streamline treasury management and facilitate fiscal decentralization.

[7] Before it was eliminated, in mid-2005 the secondary reserve requirement was reduced from 35 to 15 percent.

- **The government's three-year public financial management (PFM) plan has been enhanced.** The new PFM plan has prioritized and sequenced the reform agenda. The government is also working on costing activities to identify funding gaps.

15. **Another continuing reform involves the pricing mechanisms for selected state enterprises.** Since April 2006 the authorities are reviewing petroleum product prices monthly instead of quarterly to reduce lead-lag effects on prices and thus on the receipts of the Tema Oil Refinery (TOR). Moreover, the authorities have moved from ad valorem to specific taxation on petroleum products, effective August 2006. Also, the government has announced that it will sell 70 percent of TOR—35 percent to a strategic investor and 35 percent through the Ghana Stock Exchange.

16. **Experience with measures to ameliorate the adverse impact of removing petroleum price subsidies on the welfare of the poor has been positive**. Removal of fees for primary school has led to increased enrollment from poor families; the enhanced public transportation system for vulnerable urban groups is functioning well; and rural electrification has been extended to mitigate the pressure to substitute fuel wood for kerosene. All these mitigating measures have helped soften the public resistance to the removal of petroleum product price subsidies.

17. **However, the tariff increases of 10 percent for electricity and 21 percent for water that were planned for May 2006 have not yet been put into effect**. In the July supplementary budget, Parliament approved investments that will start to improve the quality of these services, including addressing the recent electricity shortage. The authorities expect that, with their planned actions to address the energy crisis, improvement in the quality of service would become discernible in the next couple of months, paving the way for the higher tariffs to be passed on to consumers. Meanwhile, the government is compensating the utility companies for the delay in raising tariffs, using debt write-offs and savings from elsewhere in the budget.[8] Nonetheless, the mission stressed the need to implement the planned tariff increases in a timely manner. The Public Utility Regulatory Committee (PURC) is revising the formula for utility tariffs to take into account not only fluctuations in inflation and the cedi-U.S. dollar exchange rate but also increases in utility company operating costs. Also, as of September 2006 the PURC has abolished quarterly tariff adjustments, allowing the utility companies to apply for tariff adjustments as needed.

IV. Use of MDRI Resources

18. **The discussion of MDRI debt relief centered on the modalities and timing of transfers from the central bank to the government, use of the resources, and the mechanism that has been instituted to track their use.** A subaccount of the government's consolidated account with the central bank has been established to receive and disburse MDRI resources. The central bank is transferring amounts equivalent to the IMF MDRI debt

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[8] So far, the government has paid about US$22 million to the utilities by offsetting the debt they owe to the government.

relief in installments, starting with an installment of US$200 million in June 2006. In line with the Ghana poverty reduction strategy, these resources are being used primarily on investment in physical infrastructure (particularly in rural areas), social sector spending (mainly education and health), and governance (courts and police). The July 2006 supplementary budget included total MDRI resources of US$229 million (1.8 percent of GDP) released in 2006 (from the IMF, World Bank, and African Development Fund) to be used in almost equal amounts for poverty-related expenditure and growth-enhancing investment. The amounts for 2007-08 will be incorporated into the annual budgets.

V. Medium-Term Outlook and External Debt Sustainability

A. Medium-Term Outlook

19. **The staff medium-term baseline scenario, based on current investment levels and envisaged resources from donors, is broadly similar to that presented in the last staff report**. The scenario was updated to reflect new information on 2006 results to date, projected resources from donors following the June 2006 Consultative Group meeting, and the outlook for the global economy. In this scenario, real GDP growth is projected to increase marginally from the current rate of 6 percent in 2006 to an annual average of 6.3 percent in 2007–11 (Table 10). Also, the scenario assumes limited investment in physical infrastructure, an improved environment for private sector activities, and a more skilled labor force over the medium term. The baseline scenario is conservative, assuming no step-up in the reform agenda, nor significant changes in the level or efficiency of government investment (in particular in key infrastructure such as energy, roads, railways, water and sanitation, etc). However, even this rate of growth could be significantly lower at an average of 5 percent a year if the current energy shortage is not addressed in a timely manner.

20. **Other macroeconomic objectives for the medium term are to further reduce inflation and to preserve a relatively strong external position**. Inflation is targeted to decline to single digit by the end of 2007 and to trend toward that in the major trading partner countries in the medium term. While the external current account deficit would widen, capital inflows are expected to be sufficient to finance the deficit and permit accumulation of enough international reserves that by 2011 import coverage is close to four months (Table 11). Export and import volumes are expected to grow at rates similar to the historical averages.

21. **To achieve these medium-term objectives, macroeconomic and structural policy strategies will remain broadly as outlined in the last staff report**:[9] Fiscal policy will be directed to providing enduring space for productive and propoor spending. The domestic debt-to-GDP ratio will remain the nominal fiscal anchor; low-priority current expenditure will be reduced to free up resources; and productive capital spending on health, education, and infrastructure will be increased to help Ghana move toward meeting the MDGs. Monetary policy will be directed to containing inflation and building international reserves.

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[9] EBS/06/72, May 26, 2006.

Reliance will be on productivity gains to maintain and improve competitiveness, as the authorities do not intend to undertake any actions—apart from short-term smoothing and achievement of the international reserve target—that would inhibit changes to the exchange rate for the cedi in response to market forces. The medium-term projections also assume the continuation of structural reforms to strengthen institutional capacity and improve the environment for private investment.

22. **The authorities stressed the need for a medium-term macroeconomic framework in line with their scaled-up investment plan and its expected contribution to accelerating growth.** In this case (alternative scenario A in Table 12), real GDP growth would be significantly higher, averaging around 7½ percent a year in 2007-11 on the strength of the policies outlined above supplemented by significant increases in investment in physical infrastructure, and measures to enhance the general efficiency of public capital and investment. The authorities believe a substantial increase in investment will be required to attain the higher growth path necessary for achieving the targets of the Ghana Poverty Reduction Strategy (GPRS II) and the Millennium Development Goals (MDGs) and for putting the country on a steady path toward middle-income status (Table 9). The authorities expect that the higher investment will lead to significant increases in growth with a gestation period of about two years. However, the projects in the investment plan will be reassessed to ascertain their feasibility, marginal productivity and projected overall impact on growth, with the assistance of the World Bank and relevant donor partners.[10]

23. **The scaled-up investment plan is consistent with and supplements the GPRS II.** It concentrates on upgrading the country's infrastructure—transport, utilities, and energy—and providing an environment conducive to private sector–led growth. To accelerate growth, the authorities plan to work first to enhance prospects in agriculture, given its importance for employment and output; raise the productivity of the labor force; and increase access to and lower the cost of capital. Also, recent improvements in PFM will help safeguard the quality of spending. Moreover, a high-level committee has already identified the priority high-return investment projects—which are invariably with high import content—that would alleviate supply bottlenecks, improve productivity and crowd in private sector investment. The authorities recognize that their scaled up investment plan may put some pressure on resources but consider nevertheless that the medium-term inflation targets would remain achievable. Nonetheless, they stressed that appropriate vigilance will be maintained to respond in a timely manner to any adverse development as the investment plan is executed—including possible impact on the real effective exchange rate, inflation and general macroeconomic stability.

24. **However, additional resources would be needed to finance the scaled-up investment plan and to stimulate an acceleration in growth rates**. The authorities estimate that they will need about US$470 million more a year (about 3 percent of 2006 GDP) for the next five years. They suggested that these resources might come from capital

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[10] This is an area where intensive work is expected over the next several months, involving the government, the World Bank and donors.

budget reprioritization, budget savings, some scaling up of concessional assistance, additional domestic financing consistent with the fiscal anchor of holding firm on the domestic debt-to-GDP ratio, and—as a last resort in extremely important cases—nonconcessional borrowing to finance specific projects with high returns. Thus, any access to nonconcessional borrowing is expected to be significantly less than the identified gap of US$470 million a year. Also, the authorities pointed out that some of the projects in the investment plan will be hived off to the private sector and public enterprises where they are commercially viable, thereby reducing the resources the government will need. In this regard, the mission cautioned on potential risks stemming from government guarantees and public-private partnerships.

25. **The main risk to the authorities' medium-term is that large borrowings are undertaken to finance the authorities' scaled-up investment, but the increase in investment does not push the growth trajectory high enough**. If the productivity of planned government investment is significantly lower than envisaged—because of, e.g., lower than expected returns on projects, lack of improvement in investment allocation and efficiency, or implementation difficulties—the projected acceleration in medium-term growth may not materialize. Hence, alternative scenario B in Table 13 assumes that the increase in government investment is the same as in the authorities' alternative scenario A, but that real GDP growth is relatively low at the historical rate of the baseline scenario. Export volume growth would then be lower than in the authorities' high growth alternative scenario, though import volume growth would be only slightly lower. Because international reserves would accumulate much more slowly, import cover would hold at about three months.

B. External Debt Sustainability

26. **This section summarizes the illustrative debt sustainability implications of the three medium-term scenarios discussed with the authorities**: the baseline scenario of continuation of current policies; a scenario with nonconcessional borrowing for scaled up investment in high-return projects that is favored by the authorities; and a scenario with nonconcessional borrowing with low growth, where the productivity of investment is much lower than in the authorities' high growth scenario (Box 1).[11]

27. **In the baseline scenario with continuation of current investment levels, Ghana has only a moderate risk of debt distress.** Using cautious long-term GDP and export growth projections, the NPVs of debt-to-GDP and debt-to-export ratios do not suggest solvency problems. A comfortable debt service-to-export ratio throughout the projected period does not indicate the possibility of difficulties in making future debt service payments. In fact, results of the baseline scenario illustrate that following HIPC and MDRI, Ghana's capacity to take on new debt has increased. However, the stress tests for this scenario indicate that debt sustainability would deteriorate in the face of various negative macroeconomic shocks, particularly a lower growth path reflecting potential negative impact

[11] A set of illustrative debt sustainability scenarios is presented in Appendix V.

of the current energy shortage, but the overall risk of debt distress for the baseline scenario is moderate.

28. **Alternative scenarios with nonconcessional borrowing indicate that the risks to debt sustainability depend crucially on the productivity of the additional investment.** In a scenario with high productivity and a strengthened policy environment (Alternative A) that the authorities prefer, the risk of debt distress remains moderate, although higher than in the baseline. If the additional investment is not translated into higher GDP and export growth, the risks of debt distress increase significantly. This underscores the importance of a cautious borrowing strategy, backed by prudent macroeconomic policies and key structural reforms to reduce the country's vulnerability to shocks over the longer term—including through diversifying production and exports.

29. **The authorities stated that, to keep debt sustainable, Ghana will be very cautious about contracting new loans, carefully monitor debt accumulation, and use all resources productively including limited nonconcessional borrowing to finance high-return projects**. To that end they agreed that it will be important to continue improving the database for and management of external debt. While recent progress has been impressive,[12] the authorities were also interested in further building their capacity to carry out debt forecasting and strengthen debt sustainability analyses by using the joint IMF/World Bank low-income country debt sustainability analysis framework.

[12] In terms of improved loan administration, enhanced institutional and legal framework for debt management, and accurate and timely reporting of debt data. Moreover, the authorities have conducted an independent debt sustainability analysis, using a HIPC framework.

Box 1. Summary of Debt Sustainability Scenarios

In the baseline scenario, Ghana's risk of debt distress is moderate. This scenario assumes continuation of current investment levels and an average official financing between 2007 and 2011 of 4 percent of GDP a year. In this scenario, all the external debt burden indicators are below their policy-dependent thresholds during the projection period. The NPV of debt-to-exports ratio threshold is breached at the end of the period under stress tests reflecting possible macroeconomic shocks. This scenario clearly illustrates that following MDRI and HIPC, Ghana's capacity to take on new debt has increased. The debt-service-to-exports ratio, which reflects the terms of the borrowing, indicates that there is room for some of the borrowing to be made on nonconcessional terms, provided it is for key projects that would stimulate higher growth of GDP and exports.

Ghana: External PPG Debt Burden Indicators[1]

	Indicators		Thresholds[2]
	2006	2016	
NPV of debt-to-GDP	19.8	23.3	50
NPV of debt-to-exports	50.5	110.2	200
Debt service to exports	11.7	6.1	25

Source: IMF staff estimates.
[1] Debt burden indicators are expressed in percent.
[2] These are policy-dependent thresholds that are applied to low-income countries in the joint IMF/WB debt sustainability framework.

According to the analysis, **Ghana's debt sustainability would remain moderate** under Alternative Scenario A (commercial borrowing with high growth), although the risk to debt sustainability would be higher that in the baseline scenario. However, if nonconcessional borrowing were not accompanied by any increase in the growth rate compared to the baseline (Alternative Scenario B), debt sustainability would deteriorate significantly. In that case, Ghana would be at high risk of debt distress as the debt service ratio would approach its threshold towards the end of the period. All stress tests and historical scenarios would also exceed the thresholds. Thus under this "high risk" scenario, debt sustainability remains highly vulnerable to even small shocks.

VI. Staff Appraisal

30. **Ghana's economic performance under the three-year program that started in May 2003 has been quite satisfactory, even though early slippages had to be corrected to keep the program on course**. Performance continued to be good through the first half of 2006. Real GDP is estimated to be growing at around 6 percent; inflation—despite sharp adjustments in petroleum prices—was reduced from about 15 percent at the end of 2005 to 10½ percent in June 2006; and gross international reserves rose appreciably. The prospects for 2006 as a whole are for economic activity to continue at the current pace and for further macroeconomic consolidation.

31. **Satisfactory policy implementation supported good economic performance in the first half of 2006**. The authorities have continued to consolidate the fiscal position and to control liquidity expansion. Reforms are beginning to yield the desired results in (i) the

public enterprise sector, enhancing cost recovery—particularly in the domestic pricing of petroleum products; (ii) the financial sector, deepening the intermediation process; (iii) public finance management, increasing the transparency and efficient allocation of resources; (iv) the civil service, laying the groundwork for the delivery of quality services more efficiently; and (v) public debt management, forecasting debt and analyzing loan proposals.

32. **All but one of the quantitative and structural performance criteria and benchmark for the end of June 2006 were met, and some were surpassed**. Because the ceiling on the net domestic assets of the Bank of Ghana was exceeded by a relatively small margin and only for technical reasons, staff (i) supports the authorities' request for a waiver of the nonobservance of this criterion and (ii) recommends completion of the sixth and final review under the PRGF arrangement.

33. **Ghana would like to accelerate economic growth to 8½ percent a year in the next few years**. The authorities have prepared a scaled-up investment plan within the context of a relatively stable macroeconomic framework. The main challenge for Ghana now is to find the resources it needs at terms that do not jeopardize its newfound debt sustainability while continuing to improve the policy environment and undertaking investments that are sufficiently productive to accelerate growth.

34. **There are two main risks to the authorities' medium-term development strategy**. First, if investment is incomplete because the required resources are not obtained, economic growth will be slower. The macroeconomic picture would nevertheless still be broadly satisfactory and external debt would remain relatively sustainable even with certain shocks. Second, if the productivity of investment is not as robust as the authorities envisage, the growth rate would be lower and the debt acquired to finance the investment would render debt sustainability extremely vulnerable to even small shocks. Therefore, in the aftermath of HIPC, MDRI, and bilateral debt relief, the staff welcomes the authorities' commitment to be highly circumspect in contracting new debt.

VII. Proposed Decision

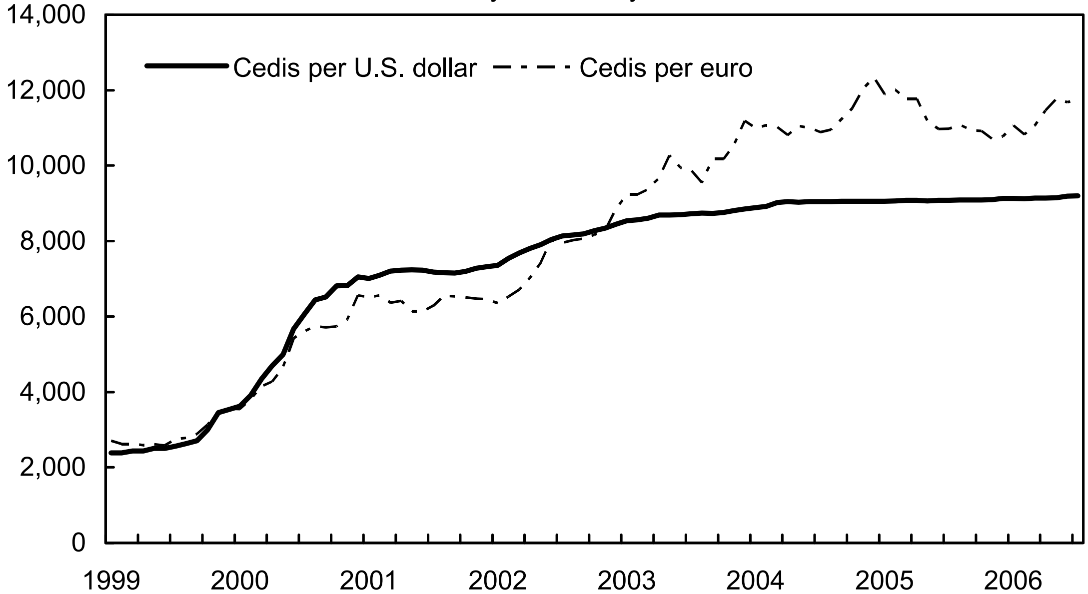
The following decision, which may be adopted by a majority of the votes cast, is proposed for adoption by the Executive Board:

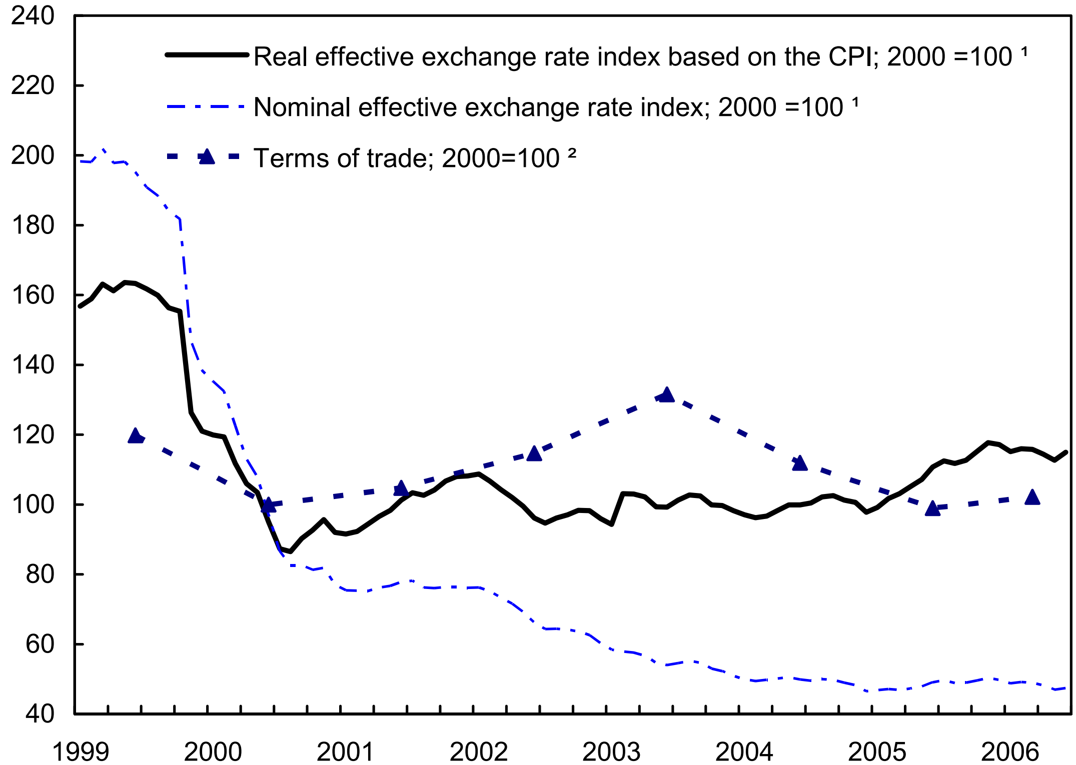
1. Ghana has consulted with the Fund in accordance with paragraph 2(I)(E)(c) of the three-year arrangement for Ghana under the Poverty Reduction and Growth Facility (PRGF) (EBS/03/42, Supplement 2, 5/14/03), as amended, to review program implementation.

2. The letter from the Minister of Finance and Economic Planning and the Governor of the Bank of Ghana dated October 18, 2006 (the "Letter") shall be attached to the three-year PRGF Arrangement for Ghana, and the letter from the Minister of Finance and Economic Planning and the Governor of the Bank of Ghana dated March 31, 2003, as amended, shall be read as further supplemented and modified by the Letter.

3. The Fund decides that the sixth review contemplated in paragraph 2(I)(E)(c) of the arrangement is completed, and that Ghana may request the disbursement of the seventh loan referred to in paragraph 1(c)(vii) of the arrangement, notwithstanding the nonobservance of the quantitative performance criterion on the net domestic assets of the Bank of Ghana for end-June 2006 set out in paragraph 2(I)(E)(a)(ii) of the arrangement, on the condition that the information provided by Ghana on the observance of the performance criterion is accurate.

Figure 1. Ghana: Exchange Rate and Terms of Trade Developments, January 1999—July 2006



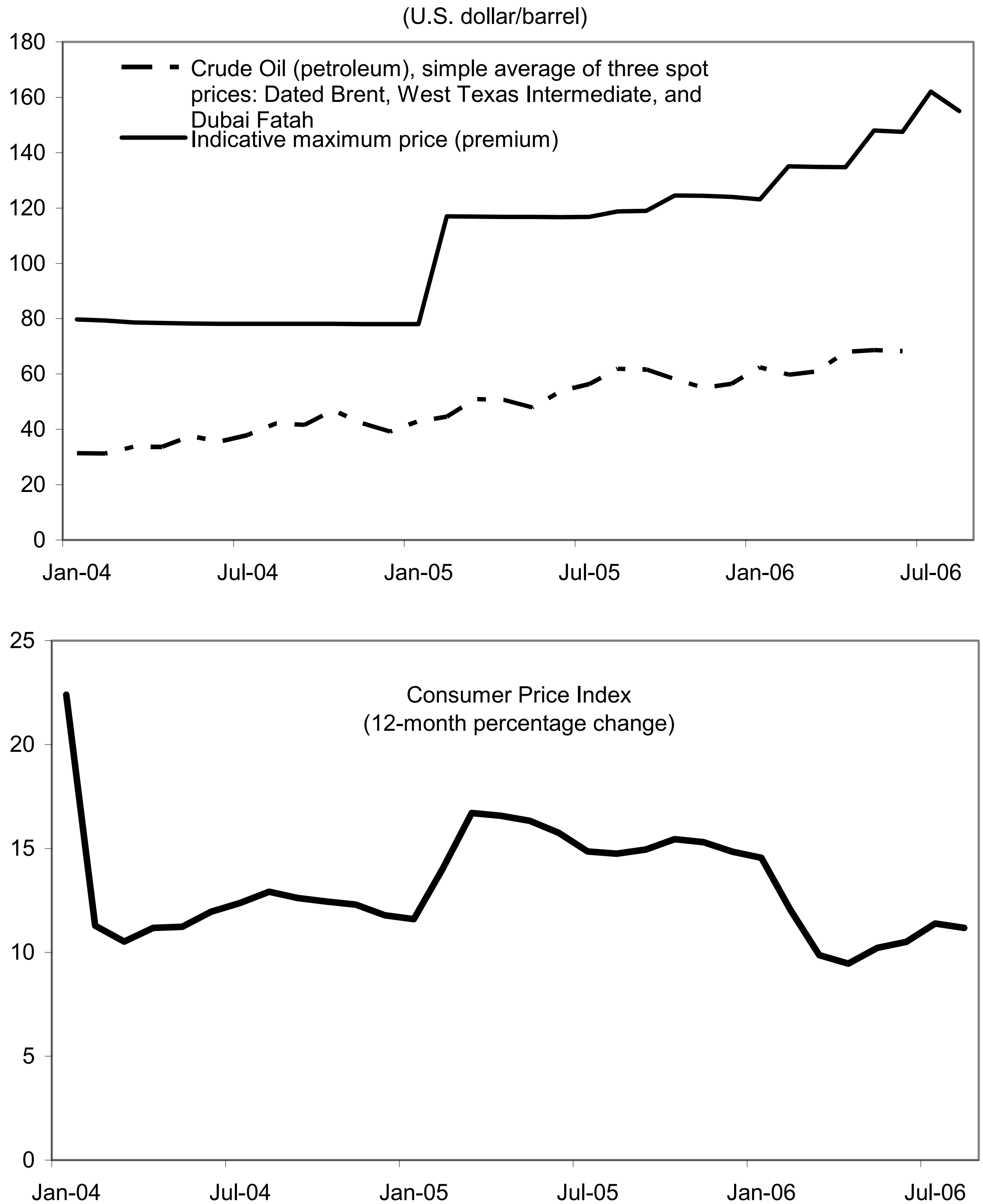


Sources: IMF, Information Notice System; and IMF staff estimates.

[1] An increase in the index denotes an appreciation.

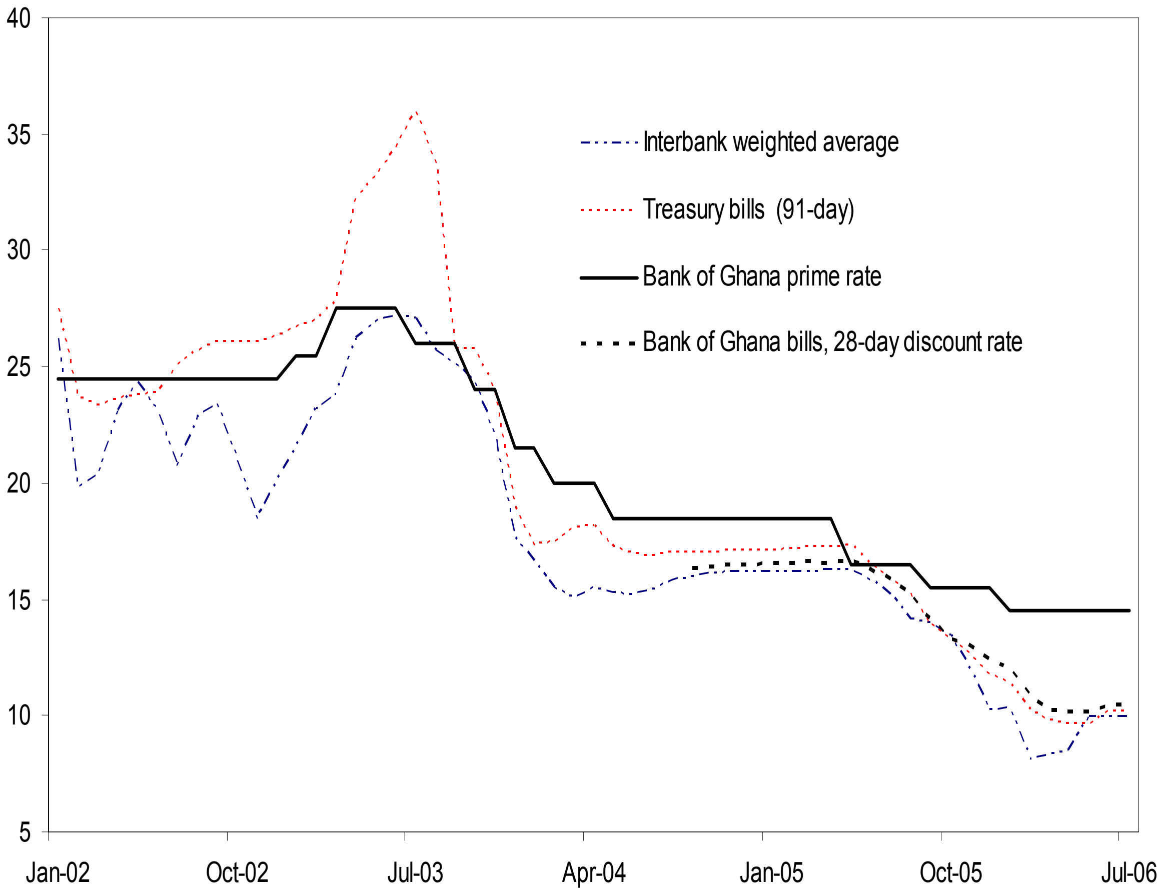
[2] An increase in the index denotes an improvement in the terms of trade.

Figure 2. Ghana: Consumer and Oil Prices, January 2004-August 2006



Sources: IMF, World Economic Outlook; and Ghanaian authorities.

Figure 3. Ghana: Interest Rate Developments, January 2002–July 2006
(Percent)



Source: Ghanaian authorities.

Table 1. Ghana: Selected Economic and Financial Indicators, 2003-07

	2003	2004	2005	2006		2007
				Prog.[1]	Proj.	Proj.
	(Annual percentage change, unless otherwise specified)					
National accounts and prices						
Real GDP	5.2	5.6	5.9	6.0	6.0	6.0
Real GDP per capita	2.6	3.0	3.2	3.3	3.4	3.3
Nominal GDP	35.4	20.8	21.7	15.8	21.7	16.0
GDP deflator	28.6	14.4	15.0	9.3	14.8	9.5
Consumer price index (annual average)	26.7	12.6	15.1	8.8	10.8	8.8
Consumer price index (end of period)	23.6	11.8	14.8	8.3	10.4	8.3
External sector						
Exports, f.o.b.	20.1	12.7	0.6	18.5	31.5	5.9
Imports, f.o.b.	20.0	31.9	24.4	10.4	19.3	8.1
Export volume	-6.8	14.5	-0.9	12.8	12.1	2.6
Import volume	6.8	14.0	7.4	8.5	8.8	3.8
Terms of trade	14.8	-15.0	-12.3	3.2	7.0	-0.9
Nominal effective exchange rate (end of period)	-15.5	-8.7	7.1	…	…	…
Real effective exchange rate (end of period)	2.4	-0.5	19.8	…	…	…
Cedis per U.S. dollar (avg.)	8,677	9,005	9,073	…	…	…
Cedis per U.S. dollar (end of period)	8,852	9,054	9,131	…	…	…
Government operations						
Domestic revenue (excluding grants)	56.2	38.2	21.9	14.1	14.2	17.8
Total expenditure	49.3	39.7	12.5	21.6	29.9	10.1
Current expenditure	34.4	27.2	9.1	15.5	19.9	5.5
Capital expenditure and net lending	97.8	67.2	18.1	15.6	45.5	16.1
Money and credit						
Net domestic assets [2]	7.2	1.0	7.0	-2.9	3.3	3.0
Credit to central government [2]	-8.2	30.9	3.3	-3.5	-3.8	-0.5
Credit to public enterprises [2]	11.2	6.6	-0.3	1.1	1.5	1.1
Credit to the private sector [2]	16.4	10.0	23.1	16.4	15.2	21.2
Broad money (excluding foreign currency deposits)	40.9	26.4	13.7	18.3	18.3	18.4
Reserve money	33.4	18.5	11.2	17.5	17.2	17.1
Velocity (GDP/average broad money)	4.7	4.3	4.3	4.3	4.2	4.1
Treasury bill yield 91 days (percent; end of period)	18.1	16.4	11.5	…	…	…
Prime rate (Bank of Ghana; percent; end of period)	25.7	19.1	15.5	…	…	…

Table 1. Ghana: Selected Economic and Financial Indicators, 2003-07 (concluded)

	2003	2004	2005	2006		2007
				Prog.[1]	Proj.	Proj.
	(Percent of GDP, unless otherwise specified)					
Investment and saving						
Gross investment	22.9	28.4	29.9	30.0	32.3	32.4
Private [3]	14.0	16.0	17.9	17.6	18.0	18.0
Central government	8.9	12.4	12.0	12.4	14.3	14.4
Gross national saving	20.7	25.7	22.8	23.5	27.1	27.7
Private [3]	14.5	14.9	12.0	11.3	15.4	16.8
Central government	6.2	10.8	10.8	12.2	11.7	10.9
Government operations						
Total revenue	20.8	23.8	23.8	23.5	22.3	22.7
Grants	4.7	6.4	5.2	5.7	6.3	5.2
Total expenditure	28.8	33.3	30.7	32.4	32.8	31.1
Overall balance (including grants)	-4.4	-3.6	-3.0	-3.6	-4.6	-3.4
Domestic primary balance	2.3	0.7	3.4	0.8	-1.3	0.8
Divestiture receipts	0.6	0.4	0.2	0.5	0.6	0.0
Net domestic financing	0.6	0.1	-1.7	0.1	0.1	-0.1
Net foreign financing	3.3	3.3	3.7	3.5	4.4	3.5
Discrepancy/resource gap	-0.2	-0.1	0.7	-0.5	-0.5	0.0
NPV of total government debt [4]	89.7	50.4	32.0	20.1	28.2	27.4
Domestic debt (net)	18.3	15.2	10.8	8.7	8.3	7.0
External debt	71.5	35.2	21.3	11.4	19.9	20.4
External sector						
Current account balance (after grants)	1.7	-2.7	-7.0	-6.5	-5.3	-4.7
NPV of external debt outstanding						
Percent of exports of goods and services [4]	175.7	89.6	59.0	28.8	50.5	56.6
(percent of government revenue) [4]	351.0	148.9	90.0	52.3	89.1	91.2
External debt service due, including to the Fund	5.9	6.4	5.1	3.8	4.6	1.6
Percent of exports of goods and services	14.5	16.2	14.2	9.5	11.8	4.3
Percent of government revenue	23.2	21.2	17.7	12.9	16.1	5.6
	(Millions of U.S. dollars, unless otherwise specified)					
Current account balance (after grants)	127	-236	-755	-738	-683	-693
Overall balance of payments	506	-13	309	406	403	337
Change in external arrears (decrease -)	0	0	0	0	0	0
Gross international reserves (end of period)	1,427	1,816	1,951	2,084	2,090	2,427
Months of imports of goods and services	3.2	3.3	2.8	3.2	2.9	3.2
Petroleum price (U.S. dollars per barrel)	28.9	37.8	53.4	...	69.2	75.5
Trading partners' real domestic demand (percentage change)	1.7	2.5	2.2	...	2.7	2.4
Total donor support	681	837	971	...	1212	1313
Percent of GDP	8.9	9.4	9.1	...	9.4	8.9
Nominal GDP (billions of cedis)	66,158	79,888	97,261	112,320	118,404	137,312
(Millions of U.S. dollars)	7,624	8,872	10,720	...	...	...
Domestic debt (net; billions of cedis)	12,089	12,133	10,461	9,799	9,799	9,661

Sources: Data provided by Ghanaian authorities; and staff estimates and projections.

[1] EBS/06/72.

[2] Percent of broad money at the beginning of the period.

[3] Including public enterprises and errors and omissions.

[4] From 2005 onwards, figures are based on the new low-income country debt sustainability analysis framework.

Table 2. Ghana: Balance of Payments, 2002-06
(Millions of U.S. dollars; unless otherwise specified)

	2002	2003	2004	2005	2006 [1]			
					June		Dec	
					Prog.	Act.	Prog.	Proj.
Exports f.o.b.	2,057	2,471	2,785	2,802	1,660	1,859	3,319	3,684
Cocoa beans and cocoa products	463	818	1,071	908	483	623	966	1,002
Gold	689	830	840	946	616	614	1,232	1,430
Timber and timber products	182	174	212	227	120	103	239	207
Others	723	649	662	722	441	518	883	1,046
Imports, f.o.b.	-2,714	-3,259	-4,297	-5,345	-2,950	-2,926	-5,900	-6,377
Non-oil	-2,206	-2,696	-3,522	-4,218	-2,333	-2,271	-4,665	-4,803
Oil	-508	-563	-775	-1,127	-617	-655	-1,234	-1,575
Trade balance	-657	-788	-1,513	-2,543	-1,290	-1,067	-2,580	-2,693
Services (net)	-214	-280	-554	-329	-193	-364	-386	-564
Of which: interest payments [1]	-124	-110	-167	-103	-37	-47	-74	-71
Private transfers (net)	680	801	1,287	1,550	845	903	1,689	2,094
Current account balance (before grants),	-190	-266	-779	-1,322	-638	-529	-1,277	-1,163
Official transfers (net)	221	393	544	567	269	273	538	480
Current account balance (after grants),	30	127	-236	-755	-369	-256	-738	-683
Capital account	-65	161	384	868	774	212	1,118	1,120
Official capital (net)	-115	86	52	309	585	34	740	763
Inflows [2]	195	370	402	500	4,862	3,851	5,120	4,580
Amortization [1]	-309	-284	-350	-191	-4,278	-3,817	-4,380	-3,817
Private capital	50	75	332	559	189	179	379	357
Other capital and errors and omissions	75	218	-162	196	15	401	26	-34
Of which: errors and omissions	33	102	21	62	0	0	0	0
Overall balance	40	506	-13	309	420	357	406	403
Change in arrears	-61	0	0	0	0	0	0	0
Financing	21	-506	12	-309	-420	-357	-406	-403
Debt deferral	-35	-35	-35	0	0	0	-406	0
Net international reserves (- is increase) [1]	-158	-679	-184	-309	-420	-357	-406	-403
Of which: use of Fund credit	54	53	-5	-15	-304	-304	-266	-265
Disbursements (PRGF)	68	74	39	39	77	77	115	116
Repayments (PRGF)	-14	-21	-44	-54	-381	-381	-381	-381
Exceptional financing	214	208	231	0	0	0	0	0
Memorandum items:								
Current account balance (in percent of GDP)								
Excluding official transfers	-3.1	-3.5	-8.8	-12.3	...	...	-11.3	-9.0
Including official transfers	0.5	1.7	-2.7	-7.0	...	...	-6.5	-5.3
Gross international reserves (US$ millions)	632	1,427	1,816	1,951	1,990	2,017	2,084	2,090
Months of imports of goods and services	1.9	3.2	3.3	2.8	...	...	3.2	2.9
Cocoa exports								
Volume (thousands of tons)	367	403	665	580	...	...	595	630
Price (US$ per ton)	1,266	1,950	1,589	1,524	...	...	1,600	1,550
External debt service paid								
Percent of exports of GNFS	10.2	5.2	5.6	2.4	...	...	1.7	11.7
Percent of government revenue	24.1	10.2	9.2	8.9	...	...	2.9	20.6
MDRI debt-service savings	…	…	…	…	21.3	21.3	76.6	76.6
Of which: on obligations to the IMF	…	…	…	…	21.3	21.3	42.6	42.6
on obligations to the World Bank	…	…	…	…	…	…	25.0	25.0
on obligations to the African Development Bank	…	…	…	…	…	…	9.0	9.0

Sources: Bank of Ghana; and IMF staff estimates and projections.

[1] Cumulative from the beginning of the year.

[2] Including the capital transfer in 2006 of MDRI debt relief from the Fund, IDA and the African Development Fund.

[3] Debt service to bilateral creditors is net of relief from 2005 onward.

Table 3. Ghana: Summary of Central Government Budgetary Operations, 2002-06 [1]

	2002	2003	2004	2005	2006			
					Jan.- Jun.		Year	
					Prog [2]	Prel.	Prog [2, 3]	Proj.
	(Billions of cedis)							
Total revenue and grants	10,333	16,862	24,073	28,256	13,081	14,564	32,759	33,866
Revenue	8,800	13,743	18,994	23,156	10,444	10,761	26,411	26,439
Grants	1,533	3,119	5,080	5,100	2,637	3,803	6,348	7,427
Total expenditure	12,752	19,035	26,584	29,895	16,903	15,627	36,362	38,827
Recurrent expenditure	9,762	13,122	16,696	18,221	10,123	10,269	21,039	21,847
Noninterest	6,763	9,030	13,223	14,681	8,463	8,569	17,626	18,419
Wages and salaries	4,142	5,567	6,988	8,243	5,387	4,987	9,990	9,990
TOR subsidy and other petroleum-related subsidy [4]	0	156	1,772	412	0	527	805	1,332
Other	2,621	3,308	4,463	6,025	3,076	3,054	6,831	7,096
Interest [5]	3,000	4,092	3,473	3,540	1,660	1,700	3,413	3,428
Capital expenditure (total)	2,990	5,912	9,888	11,673	6,780	5,358	15,323	16,980
Of which: domestic	1,338	3,015	5,077	5,727	3,544	3,139	7,235	8,912
foreign	1,652	2,897	4,812	5,946	3,236	2,219	6,258	8,069
Arrears clearance and VAT refunds	-892	-711	-326	-451	-215	-349	-447	-447
Payments in 2005 due to previous commitments	...	...	...	-780	...	...	...	
Overall balance	-3,312	-2,884	-2,836	-2,869	-4,037	-1,412	-4,050	-5,408
Discrepancy between above- and below-line data	547	-138	-104	637	-579	-1,476	-579	-579
Total financing	2,765	3,022	2,941	2,233	4,616	2,887	4,629	5,987
Divestiture receipts	5	421	286	233	0	8	584	673
Foreign (net) [5]	463	2,202	2,611	3,640	2,525	2,429	3,902	5,170
Of which: Exceptional financing	1,241	1,804	1,065	1,193	994	567	1,731	1,731
Domestic (net)	2,297	399	43	-1,640	2,091	450	144	144
Of which: new TOR bonds [6]	...	...	...	...	...	...	805	805
Net savings due to inflation-indexed bonds	197	481	-349	-99	-79	7	-71	-71
Memorandum items:								
Total enhanced HIPC relief	759	1,163	2,117	2,155	1,153	942	2,280	2,280
Total poverty spending (GPRS definition)	2,330	4,279	6,123	8,256	4,242	4,613	11,187	11,024
Domestic primary balance [7]	1,192	1,510	535	3,268	-2,183	-2,501	871	-1,571
Domestic debt (net) [8]	11,690	12,089	12,133	10,461	12,552	10,911	9,799	9,799

Table 3. Ghana: Summary of Central Government Budgetary Operations, 2002-06 (concluded) [1]

	2002	2003	2004	2005	2006			
					Jan.- Jun.		Year	
					Prog [2]	Prel.	Prog [2, 3]	Proj.
	(Percent of GDP)							
Total revenue and grants	21.1	25.5	30.1	29.1	11.6	13.0	29.2	28.6
Revenue	18.0	20.8	23.8	23.8	9.3	9.6	23.5	22.3
Grants	3.1	4.7	6.4	5.2	2.3	3.4	5.7	6.3
Total expenditure	26.1	28.8	33.3	30.7	15.0	13.9	32.4	32.8
Recurrent expenditure	20.0	19.8	20.9	18.7	9.0	9.1	18.7	18.5
Noninterest	13.8	13.6	16.6	15.1	7.5	7.6	15.7	15.6
Wages and salaries	8.5	8.4	8.7	8.5	4.8	4.4	8.9	8.4
TOR subsidy and other petroleum-related subsidy [4]	0.0	0.2	2.2	0.4	0.0	0.5	0.7	1.1
Other	5.4	5.0	5.6	6.2	2.7	2.7	6.1	6.0
Interest [5]	6.1	6.2	4.3	3.6	1.5	1.5	3.0	2.9
Capital expenditure (total)	6.1	8.9	12.4	12.0	6.0	4.8	13.6	14.3
Of which : domestic	2.7	4.6	6.4	5.9	3.2	2.8	6.4	7.5
Foreign	3.4	4.4	6.0	6.1	2.9	2.0	5.6	6.8
Arrears clearance and VAT refunds	-1.8	-1.1	-0.4	-0.5	-0.2	-0.3	-0.4	-0.4
Payments in 2005 due to previous commitments	…	…	…	-0.8	…	…	…	…
Overall balance	-6.8	-4.4	-3.6	-3.0	-3.6	-1.3	-3.6	-4.6
Discrepancy between above- and below-line data	1.1	-0.2	-0.1	0.7	-0.5	-1.3	-0.5	-0.5
Total financing	5.7	4.6	3.7	2.3	4.1	2.6	4.1	5.1
Divestiture receipts	0.0	0.6	0.4	0.2	0.0	0.0	0.5	0.6
Foreign (net) [5]	0.9	3.3	3.3	3.7	2.2	2.2	3.5	4.4
Of which: Exceptional financing	2.5	2.7	1.3	1.2	0.9	0.5	1.5	1.5
Domestic (net)	4.7	0.6	0.1	-1.7	1.9	0.4	0.1	0.1
Of which : new TOR bonds [6]	...	...	...	...	...	...	0.7	0.7
Net savings due to inflation-indexed bonds	0.4	0.7	-0.4	-0.1	-0.1	0.0	-0.1	-0.1
Memorandum items:								
Total enhanced HIPC relief	1.6	1.8	2.7	2.2	1.0	0.8	2.0	1.9
Total poverty spending (GPRS definition)	4.8	6.5	7.7	8.5	3.8	4.1	10.0	9.3
Domestic primary balance [7]	2.4	2.3	0.7	3.4	-1.9	-2.2	0.8	-1.3
Domestic debt (net) [8]	23.9	18.3	15.2	10.8	11.2	9.7	8.7	8.3

Sources: Tables 4-6; data provided by Ghanaian authorities; and IMF staff estimates and projections.

[1] Above-the-line data for domestic recurrent and capital expenditure are presented on a cash basis.
[2] See EBS/06/72.
[3] Includes MDRI debt relief from the IMF.
[4] For 2006, petroleum-related subsidy to TOR and the oil marketing companies for operations in 2005.
[5] The 2006 program includes MDRI debt relief from World Bank (US$ 25 million) and the African Development Bank (US$ 9 million).
[6] Excluding additional TOR bonds issued in July 2006 to restructure debt.
[7] Defined as total revenue plus VAT refunds (negative) and discrepancy, less noninterest and domestic capital expenditure.
[8] Excluding noninterest-bearing perpetual Bank of Ghana revaluation stocks, and bonds issued in 2004-06 for recapitalization of BoG and TOR, including deferred interest payment (cumulative) on inflation indexed bonds, and less government deposits.

Table 4. Ghana: Central Government Revenues and Grants, 2002-06

	2002	2003	2004	2005	2006			
					Jan.-Jun.		Year	
					Prog.[1]	Prel.	Prog.[1,2]	Proj.
	(Billions of cedis)							
Total revenue and grants	10,333	16,862	24,073	28,256	13,081	14,564	32,759	33,866
Revenue	8,800	13,743	18,994	23,156	10,444	10,761	26,411	26,439
Tax revenue	8,542	13,346	17,341	20,070	9,509	9,995	24,080	24,107
Direct taxes	2,790	4,057	5,287	6,540	2,609	2,746	7,162	7,162
Company	1,162	1,632	2,340	3,108	1,332	1,223	3,789	3,789
Personal	929	1,535	1,908	2,435	866	1,105	2,255	2,255
Self-employed	173	234	286	347	167	151	428	428
Other	527	657	753	650	244	267	691	691
Indirect taxes	3,757	6,128	8,245	9,416	5,001	5,104	12,059	12,087
Sales tax/value-added tax (VAT) on domestic goods	729	1,023	1,460	1,832	994	1,157	2,318	2,318
Sales tax/value-added tax (VAT) on imports	1,598	2,308	3,051	3,155	1,944	1,521	4,713	4,713
Petroleum	1,080	2,323	3,119	3,751	1,639	2,080	4,148	4,175
Other	350	474	615	678	424	346	880	880
Trade taxes	1,995	3,160	3,809	4,114	1,899	2,145	4,859	4,859
Import duties	1,669	2,391	2,878	3,499	1,734	1,817	4,446	4,446
Cocoa export duty	326	769	931	615	164	328	413	413
Nontax revenue	258	397	1,194	1,929	236	277	818	818
Fees, charges, and dividends	258	363	795	1,854	189	258	711	711
Other nontax revenue	0	34	399	75	48	19	107	107
Health levy	...	...	459	1,157	699	489	1,514	1,514
Grants	1,533	3,119	5,080	5,100	2,637	3,803	6,348	7,427
Project grants	467	1,037	2,265	2,866	1,405	965	2,508	3,002
Program grants	558	1,267	1,762	1,273	695	637	1,389	1,389
HIPC assistance (multilateral)	508	815	1,053	961	537	375	1,207	1,207
MDRI assistance from IMF	...	...	...	...	0	1,826	1,243	1,829
	(Percent of GDP)							
Total revenue and grants	21.1	25.5	30.1	29.1	11.6	13.0	29.2	28.6
Revenue	18.0	20.8	23.8	23.8	9.3	9.6	23.5	22.3
Tax revenue	17.5	20.2	21.7	20.6	8.5	8.9	21.4	20.4
Direct taxes	5.7	6.1	6.6	6.7	2.3	2.4	6.4	6.0
Company tax	2.4	2.5	2.9	3.2	1.2	1.1	3.4	3.2
Personal	1.9	2.3	2.4	2.5	0.8	1.0	2.0	1.9
Self-employed	0.4	0.4	0.4	0.4	0.1	0.1	0.4	0.4
Other direct taxes	1.1	1.0	0.9	0.7	0.2	0.2	0.6	0.6
Indirect taxes	7.7	9.3	10.3	9.7	4.5	4.5	10.7	10.2
Sales tax/value-added tax (VAT) on domestic goods	1.5	1.5	1.8	1.9	0.9	1.0	2.1	2.0
Sales tax/value-added tax (VAT) on imports	3.3	3.5	3.8	3.2	1.7	1.4	4.2	4.0
Petroleum	2.2	3.5	3.9	3.9	1.5	1.9	3.7	3.5
Other indirect taxes	0.7	0.7	0.8	0.7	0.4	0.3	0.8	0.7
Trade taxes	4.1	4.8	4.8	4.2	1.7	1.9	4.3	4.1
Import duties	3.4	3.6	3.6	3.6	1.5	1.6	4.0	3.8
Cocoa export duty	0.7	1.2	1.2	0.6	0.1	0.3	0.4	0.3
Nontax revenue	0.5	0.6	1.5	2.0	0.2	0.2	0.7	0.7
Fees, charges, and dividends	0.5	0.5	1.0	1.9	0.2	0.2	0.6	0.6
Other nontax revenue	0.0	0.1	0.5	0.1	0.0	0.0	0.1	0.1
Health levy	…	…	0.6	1.2	0.6	0.4	1.3	1.3
Grants	3.1	4.7	6.4	5.2	2.3	3.4	5.7	6.3
Project grants	1.0	1.6	2.8	2.9	1.3	0.9	2.2	2.5
Program grants	1.1	1.9	2.2	1.3	0.6	0.6	1.2	1.2
HIPC assistance (multilateral)	1.0	1.2	1.3	1.0	0.5	0.3	1.1	1.0
MDRI assistance	...	...	...	...	0.0	1.6	1.1	1.5

Sources: Ghanaian authorities; and IMF staff estimates and projections.

[1] See EBS/06/72.

[2] Includes MDRI debt relief from the IMF.

Table 5. Ghana: Central Government Expenditures, 2002-06 [1]

	2002	2003	2004	2005	2006			
					Jan.-Jun.		Year	
					Prog. [2]	Prel.	Prog. [2, 3]	Proj.
					(Billions of cedis)			
Total expenditure	12,752	19,035	26,584	29,895	16,903	15,627	36,362	38,827
Recurrent expenditure	9,762	13,122	16,696	18,221	10,123	10,269	21,039	21,847
Noninterest	6,763	9,030	13,223	14,681	8,463	8,569	17,626	18,419
Wages and salaries	4,142	5,567	6,988	8,243	5,387	4,987	9,990	9,990
Goods and services	1,452	2,063	2,737	3,105	1,467	1,823	3,141	3,086
General goods and services	1,452	2,063	2,737	2,960	1,394	1,750	2,996	2,941
Strategic oil reserves	...	...	...	145	73	73	145	145
Transfers	1,168	1,400	3,499	3,332	1,609	1,759	4,496	5,343
Pensions	271	359	515	659	305	408	706	706
Gratuities	114	201	269	328	165	224	409	409
Social security	303	412	495	631	309	410	712	712
National Health Insurance Fund	...	...	0	984	699	0	1,514	1,514
Other transfers	0	0	12	318	81	190	250	250
Utility subsidy	480	273	435	0	50	0	100	420
TOR subsidy/other petroleum-related subsidy [4]	0	156	1,772	412	0	527	805	1,332
Interest	3,000	4,092	3,473	3,540	1,660	1,700	3,413	3,428
Domestic (due)	2,212	3,268	2,545	2,689	1,108	1,300	2,427	2,427
External (due) [5]	788	824	927	851	552	399	986	1,001
Capital expenditure (total)	2,990	5,912	9,888	11,673	6,780	5,358	15,323	16,980
Of which: domestic	1,338	3,015	5,077	5,727	3,544	3,139	7,235	8,912
Cash expenditure	300	422	927	894	1,171	614	1,724	3,401
Statutory funds	863	1,871	2,280	2,886	1,562	1,575	3,688	3,688
HIPC financed expenditure	175	722	1,870	1,947	810	742	1,822	1,822
MDRI-financed expenditure					0	208	...	...
Foreign	1,652	2,897	4,812	5,946	3,236	2,219	6,258	8,069
Road arrears (clearance)	-208	-204	-71	-99	-50	-117	-100	-100
Other domestic payment arrears (clearance)	-630	-459	-199	-235	-124	-154	-247	-247
Of which: to statutory funds	-247	-97	-97	0	-48	-48	-97	-97
VAT refunds	-55	-48	-55	-117	-42	-78	-100	-100
Payments in 2005 due to previous years' commitments	...	...	...	-780	...	...	...	
					(Percent of GDP)			
Total expenditure	26.1	28.8	33.3	30.7	15.0	13.9	32.4	32.8
Recurrent expenditure	20.0	19.8	20.9	18.7	9.0	9.1	18.7	18.5
Noninterest	13.8	13.6	16.6	15.1	7.5	7.6	15.7	15.6
Wages and salaries	8.5	8.4	8.7	8.5	4.8	4.4	8.9	8.4
Goods and services	3.0	3.1	3.4	3.2	1.3	1.6	2.8	2.6
General goods and services	3.0	3.1	3.4	3.0	1.2	1.6	2.7	2.5
Strategic oil reserves	...	...	...	0.1	0.1	0.1	0.1	0.1
Transfers	2.4	2.1	4.4	3.4	1.4	1.6	4.0	4.5
Pensions	0.6	0.5	0.6	0.7	0.3	0.4	0.6	0.6
Gratuities	0.2	0.3	0.3	0.3	0.1	0.2	0.4	0.3
Social security	0.6	0.6	0.6	0.6	0.3	0.4	0.6	0.6
National Health Insurance Fund	...	...	0.0	1.0	0.6	0.0	1.3	1.3
Other transfers	0.0	0.0	0.0	0.3	0.1	0.2	0.2	0.2
Utility subsidy	1.0	0.4	0.5	0.0	0.0	0.0	0.1	0.4
TOR subsidy/other petroleum-related subsidy [4]	0.0	0.2	2.2	0.4	0.0	0.5	0.7	1.1
Interest	6.1	6.2	4.3	3.6	1.5	1.5	3.0	2.9
Domestic (due)	4.5	4.9	3.2	2.8	1.0	1.2	2.2	2.0
External (due) [5]	1.6	1.2	1.2	0.9	0.5	0.4	0.9	0.8
Capital expenditure (total)	6.1	8.9	12.4	12.0	6.0	4.8	13.6	14.3
Of which: domestic	2.7	4.6	6.4	5.9	3.2	2.8	6.4	7.5
Cash expenditure	0.6	0.6	1.2	0.9	1.0	0.5	1.5	2.9
Statutory funds	1.8	2.8	2.9	3.0	1.4	1.4	3.3	3.1
HIPC financed expenditure	0.4	1.1	2.3	2.0	0.7	0.7	1.6	1.5
MDRI-financed expenditure					0.0	0.2	...	...
Foreign	3.4	4.4	6.0	6.1	2.9	2.0	5.6	6.8
Road arrears (clearance)	-0.4	-0.3	-0.1	-0.1	0.0	-0.1	-0.1	-0.1
Other domestic payment arrears (clearance)	-1.3	-0.7	-0.2	-0.2	-0.1	-0.1	-0.2	-0.2
Of which: to statutory funds	-0.5	-0.1	-0.1	0.0	0.0	0.0	-0.1	-0.1
VAT refunds	-0.1	-0.1	-0.1	-0.1	0.0	-0.1	-0.1	-0.1
Payments in 2005 due to previous commitments	...	...	...	-0.8	...	...	...	

Sources: Ghanaian authorities; and IMF staff estimates and projections.

[1] Above-the-line data for domestic recurrent and capital expenditure are presented on a cash basis.
[2] See EBS/06/72.
[3] Includes MDRI debt relief from the IMF.
[4] For 2006, petroleum-related subsidy to TOR and the oil marketing companies for operations in 2005.
[5] The 2006 program includes MDRI debt relief from World Bank (US$25 million) and the African Development Bank (US$ 9 million).

Table 6. Ghana: Central Government Financing, 2002-06

	2002	2003	2004	2005	2006			
					Jan.-Jun.		Year	
					Prog. [1]	Prel.	Prog. [1, 2]	Proj.
				(Billions of cedis)				
Overall balance	-3,312	-2,884	-2,836	-2,869	-4,037	-1,412	-4,050	-5,408
Discrepancy between above- and below-line data	547	-138	-104	637	-579	-1,476	-579	-579
Total financing	2,765	3,022	2,941	2,233	4,616	2,887	4,629	5,987
Divestiture receipts	5	421	286	233	0	8	584	673
Foreign (net)	463	2,202	2,611	3,640	2,525	2,429	3,902	5,170
Project loans	1,185	1,860	2,547	3,080	1,832	1,254	3,749	5,066
Program loans	160	1,097	920	1,465	1,600	1,311	1,639	1,639
Amortization due [3]	-2,123	-2,559	-1,920	-2,098	-1,901	-704	-3,218	-3,266
Exceptional financing	1,241	1,804	1,065	1,193	994	567	1,731	1,731
External payment arrears	-459	0	0	0	0	0	0	0
Traditional rescheduling	1,449	1,457	0	0	378	0	657	657
HIPC assistance (nonmultilateral)	251	347	1,065	1,193	616	567	1,073	1,073
Domestic (net)	2,297	399	43	-1,640	2,091	450	144	144
Banking system	1,161	-960	1,640	338	923	-1,338	-914	-914
Nonbanks	939	878	-1,248	-1,879	1,247	1,781	324	324
New TOR bonds [4]	...	...	...	...	...	...	805	805
Net savings due to inflation-indexed bonds	197	481	-349	-99	-79	7	-71	-71
Deferred interest payment	197	481	301	67	7	7	15	15
Redemption of deferred interest payment	...	...	-650	-165	-86	0	-86	-86
				(Percent of GDP)				
Overall balance	-6.8	-4.4	-3.6	-3.0	-3.6	-1.3	-3.6	-4.6
Discrepancy between above- and below-line data	1.1	-0.2	-0.1	0.7	-0.5	-1.3	-0.5	-0.5
Total financing	5.7	4.6	3.7	2.3	4.1	2.6	4.1	5.1
Divestiture receipts	0.0	0.6	0.4	0.2	0.0	0.0	0.5	0.6
Foreign (net)	0.9	3.3	3.3	3.7	2.2	2.2	3.5	4.4
Project loans	2.4	2.8	3.2	3.2	1.6	1.1	3.3	4.3
Program loans	0.3	1.7	1.2	1.5	1.4	1.2	1.5	1.4
Amortization due [3]	-4.3	-3.9	-2.4	-2.2	-1.7	-0.6	-2.9	-2.8
Exceptional financing	2.5	2.7	1.3	1.2	0.9	0.5	1.5	1.5
External payment arrears	-0.9	0.0	0.0	0.0	0.0	0.0	0.0	0.0
Traditional rescheduling	3.0	2.2	0.0	0.0	0.3	0.0	0.6	0.6
HIPC assistance (nonmultilateral)	0.5	0.5	1.3	1.2	0.5	0.5	1.0	0.9
Domestic (net)	4.7	0.6	0.1	-1.7	1.9	0.4	0.1	0.1
Banking system	2.4	-1.5	2.1	0.3	0.8	-1.2	-0.8	-0.8
Nonbanks	1.9	1.3	-1.6	-1.9	1.1	1.6	0.3	0.3
New TOR bonds [4]	...	...	...	...	...	...	0.7	0.7
Net savings due to inflation-indexed bonds	0.4	0.7	-0.4	-0.1	-0.1	0.0	-0.1	-0.1
Deferred interest payment	0.4	0.7	0.4	0.1	0.0	0.0	0.0	0.0
Redemption of deferred interest payment	...	...	-0.8	-0.2	-0.1	0.0	-0.1	-0.1

Sources: Ghanaian authorities; and IMF staff estimates and projections.

[1] See EBS/06/72.

[2] Includes MDRI debt relief from the IMF.

[3] The 2006 program reflects MDRI debt relief from the World Bank (US$ 25 million) and the African Development Bank (US$ 9 million).

[4] Excluding additional TOR bonds issued in July 2006 to restructure debt.

Table 7. Ghana: Monetary Survey, 2002-06

	2002	2003	2004	2005	2006			
	Dec.	Dec.	Dec.	Dec.	Jun.		Dec.	
					Prog.[1]	Act.	Prog.[1]	Proj.
	(Billions of cedis, unless otherwise specified; end of period)							
Net foreign assets	6,604	10,575	14,802	16,224	21,449	19,293	23,242	19,808
(Millions of U.S. dollars)	783	1,195	1,635	1,716	2,156	2,102	2,159	2,153
Net domestic assets	8,733	10,599	11,865	14,244	11,709	14,786	12,747	16,240
Domestic credit	15,775	17,784	25,685	31,165	31,327	33,255	34,452	34,555
Claims on government (net) [2, 3]	8,402	7,433	12,569	13,257	14,502	13,020	12,665	12,343
Claims on non-government [2]	7,373	10,351	13,116	17,907	16,825	20,235	21,787	22,211
Other items, net (assets +)	-7,043	-7,185	-13,821	-16,920	-19,618	-18,469	-21,705	-18,312
Broad money (M2+) [4]	15,336	21,174	26,667	30,468	33,158	34,079	35,988	36,048
Currency	4,672	6,338	7,303	8,026	7,549	7,620	9,543	9,536
Deposits [4]	10,664	14,836	19,364	22,442	25,609	26,459	26,445	26,512
Broad money (M2) [5]	11,783	16,598	20,986	23,870	25,535	26,943	28,266	28,248
Memorandum items:	(Annual percentage change, unless otherwise specified)							
Broad money (M2)	50.7	40.9	26.4	13.7	21.9	28.6	18.3	18.3
Reserve money (RM)	42.6	33.4	18.5	11.2	19.9	19.7	17.5	17.2
Broad money (M2+)	50.5	38.1	25.9	14.3	22.3	25.7	18.3	18.3
Reserve money (RM+) [6]	48.2	34.8	17.6	6.9	17.0	17.7	17.6	17.3
Velocity (GDP/average M2+)	3.8	3.6	3.3	3.4	3.4	3.2	3.4	3.3
Velocity (GDP/average M2)	5.0	4.7	4.3	4.3	4.4	4.1	4.3	4.2
Reserve money multiplier (M2/RM)	2.04	2.16	2.30	2.35	2.67	2.83	2.37	2.38
Reserve money multiplier (M2+/RM)	2.66	2.75	2.92	3.00	3.47	3.57	3.02	3.03
Reserve money multiplier (M2+/RM+)	2.40	2.45	2.63	2.81	3.19	3.27	2.82	2.83
Bank reserves-to-deposits ratio [5]	0.1	0.1	0.1	0.1	0.1	0.1	0.1	0.1
Foreign currency deposits to total deposits	0.3	0.3	0.3	0.3	0.3	0.3	0.3	0.3
Currency-to-deposits ratio [4]	0.4	0.4	0.4	0.4	0.3	0.3	0.4	0.4
Currency/M2+ ratio	0.3	0.3	0.3	0.3	0.2	0.2	0.3	0.3
Credit to the private sector [7]	31.5	34.7	21.4	47.8	12.1	37.9	24.9	24.9
	(Millions of U.S. dollars)							
Net international reserves [8]	483	798	982	1,291	1,712	1,649	1,698	1,694

Sources: Table 8; Ghanaian authorities; and IMF staff estimates and projections.

[1] EBS/06/72.

[2] Tema Oil Refinery debt swap in December 2002 moved GC1,421 billion from credit to public enterprises to claims on government.

[3] The figure for end-June 2006 includes a new TOR bond that is excluded from the fiscal accounts.

[4] Including foreign currency deposits.

[5] Excluding foreign currency deposits.

[6] Including foreign currency reserves of deposit money banks with Bank of Ghana (BoG).

[7] Credit from deposit money banks to the private sector.

[8] Net international reserves are defined as short-term foreign assets of the BoG, minus short-term external liabilities.

Table 8. Ghana: Bank of Ghana and Deposit Money Banks, 2002-06

(Billions of cedis, unless otherwise specified; end of period)

	2002	2003	2004	2005	2006			
	Dec.	Dec.	Dec.	Dec.	Jun.		Dec.	
					Prog.[1]	Act.	Prog.[1]	Proj.
Bank of Ghana								
Net foreign assets	5,329	8,590	11,917	13,588	18,382	16,571	19,739	16,840
(Millions of U.S. dollars)	631	970	1,316	1,428	1,848	1,806	1,834	1,830
Net domestic assets	438	-897	-2,799	-3,448	-8,831	-7,037	-7,825	-4,952
Claims on government (net)	2,867	2,525	6,231	5,044	6,564	2,838	5,596	2,838
Claims on deposit money banks [2]	-375	-4,412	-7,127	-7,552	-8,070	-7,781	-8,281	-7,318
Claims on rest of the economy	271	-142	-203	-31	-31	329	-31	329
Other items, net (assets +)	-2,325	1,132	-1,701	-909	-7,295	-2,423	-5,110	-800
Reserve money	5,767	7,693	9,118	10,140	9,551	9,534	11,914	11,888
Currency outside banks	4,672	6,338	7,303	8,026	7,549	7,620	9,543	9,536
Bank reserves	1,056	1,259	1,697	2,003	1,891	1,844	2,261	2,279
Cash	273	342	371	495	398	333	553	568
Deposits	783	917	1,327	1,509	1,494	1,511	1,707	1,711
Nonbank deposits	40	96	117	110	110	70	110	70
Deposit money banks								
Net foreign assets [3]	1,275	1,985	2,885	2,636	3,068	2,723	3,502	2,969
(Millions of U.S. dollars)	151	224	319	289	308	296	325	323
Reserves	1,056	1,259	1,697	2,003	1,891	1,844	2,261	2,279
Net claims on Bank of Ghana [2]	375	4,412	7,127	7,552	8,070	7,781	8,281	7,318
Domestic credit	12,637	15,400	19,657	26,151	24,793	30,088	28,886	31,388
Claims on government (net) [4, 5]	5,535	4,907	6,338	8,213	7,938	10,182	7,068	9,505
Claims on non-government [4]	7,102	10,493	13,319	17,938	16,856	19,906	21,818	21,882
Public enterprises [4]	965	2,148	2,902	3,307	3,307	3,106	3,582	3,649
Private sector	5,980	8,052	9,778	14,455	13,538	16,657	18,059	18,048
Other items, net (assets +) [2]	-4,718	-8,317	-12,120	-16,011	-12,323	-16,046	-16,595	-17,512
Total deposits	10,625	14,740	19,247	22,332	25,499	26,389	26,335	26,442

Sources: Ghanaian authorities; and IMF staff estimates and projections.

[1] As in EBS/06/72.

[2] Includes, starting January 2003, holding of T-bills issued for monetary liquidity purposes and, starting September 2004, Bank of Ghana bills.

[3] A large part of the net foreign assets of deposit money banks for 1998-2003 was reclassified as net domestic assets to ensure consistency with the new banking supervision reporting form introduced in July 2003, which uses a residency rather than currency definition of net foreign assets.

[4] TOR debt swap moved GC1,421 billion from credit to public enterprises to claims on government in December 2002.

[5] The figure for end-June 2006 includes a new TOR bond that is excluded from the fiscal accounts.

Table 9. Ghana: Country Profile, Millennium Development Goals, 1990–2004

	1990	1995	2001	2003	2004	2015 Target	Status [1]
1. Eradicate extreme poverty and hunger.[2]							**On track**
Population below US$1 a day (percent)	…	29 [3]	45	39	…	15	On track
Poverty gap at US$1 a day (percent)	…	7 [3]	17	…	…	…	On track
Percentage share of income or consumption held by poorest 20 percent	…	…	6	…	…	…	
Prevalence of child malnutrition (percent of children under 5)	30	27	25	22	22	…	Off track
Population below minimum level of dietary energy consumption (percent)	35	17	12	…	…	…	Off track
2. Achieve universal primary education.[4]							**On track**
Net primary enrollment ratio (percent of relevant age group)	52	…	56	62	58	100	On track
Percentage of cohort reaching grade 5 (percent)	81	…	…	…	…	100	
Youth literacy rate (percent of ages 15–24)	82	87	92	…	71	…	
3. Promote gender equality.[5]							**On track**
Ratio of girls to boys in primary education (percent)	82	87	94	98	94	100	
Ratio of girls to boys in primary and secondary education (percent)	79	…	91	…	91	100	On track
Ratio of young literate females to males (percent of ages 15–24)	86	…	…	…	86	100	On track
Share of women employed in the nonagricultural sector (percent)	57	…	…	…	…	…	
Proportion of seats held by women in national parliament (percent)		8		9	9		
4. Reduce child mortality.[6]							**Off track**
Under 5 mortality rate (per 1,000)	122	110	112	…	112	42	Off track
Infant mortality rate (per 1,000 live births)	75	67	68	…	68	…	Off track
Immunization, measles (percent of children under 12 months)	61	70	81	83	83	…	
5. Improve maternal health.[7]							**Off track**
Maternal mortality rate (modeled estimate, per 100,000 live births)	740	590	540	…	…	185	Off track
Births attended by skilled health staff (percent of total)	40	44	…	…	47	…	
6. Combat HIV/AIDS, malaria, and other diseases.[8]							**On track**
Prevalence of HIV, female (percent ages 15–24)	…	…	3	4	…	…	On track
Contraceptive prevalence rate (percent of women ages 15–49)	13	20		25	25	…	
Number of children orphaned by HIV/AIDS			140,000	130,000	170,000	…	
Incidence of tuberculosis (per 100,000 people)	516		377	376	376		
Tuberculosis cases detected under DOTS (percent)	…	16	41	40	37	…	
7. Ensure environmental sustainability.[9]							**Off track**
Forest area (percent of total land area)	33	…	27	…	24	…	
Nationally protected areas (percent of total land area)	15	15	15	15	15	…	
GDP per unit of energy use (PPP $ per kg oil equivalent)	5	5	5	…	5	…	
CO2 emissions (metric tons per capita)	0.2	0.3	0.3	0.4	…	…	
Access to an improved water source (percent of population)	55	…	73	79	75	83	On track
Access to improved sanitation (percent of population)	15	…	…	…	18	…	
Access to secure tenure (percent of population)							
8. Develop a Global Partnership for Development.[10]							**On track**
Youth unemployment rate (percent of total labor force ages 15–24)	…	…	…	…	…	…	
Fixed line and mobile telephones (per 1,000 people)	3	4	24	…	93	…	
Personal computers (per 1,000 people)	0	1	3	…	5	…	

Sources: World Development Indicators database, November 2005 and September 2006,
United Nations site for MDG indicators http://mdgs.un.org/unsd/mdg/Data.aspx?cr=293) and IMF staff estimates and projections.
Note: In some cases the data are for earlier or later years than those stated.

[1] Status as reported in the Ghana Poverty Reduction Strategy 2006 and the Annual Progress Report (2005).

[2] Targets: Halve, between 1990 and 2015, the proportion of people whose income is less than $1 a day. Halve, between 1990 and 2015, the proportion of people who suffer from hunger.

[3] Data for 1994.

[4] Target: Ensure that, by 2015, children everywhere, boys and girls alike, will be able to complete a full course of primary schooling.

[5] Target: Eliminate gender disparity in primary and secondary education preferably by 2005 and to all levels of education no later than 2015.

[6] Target: Reduce by two-thirds, between 1990 and 2015, the under-5 mortality rate.

[7] Target: Reduce by three-quarters, between 1990 and 2015, the maternal mortality ratio.

[8] Targets: Have halted by 2015, and begun to reverse, the spread of HIV/AIDS. Have halted by 2015, and begun to reverse, the incidence of malaria and other major deseases.

[9] Targets: Integrate the principles of sustainable development into country policies and programs and reverse the loss of environmental resources. Halve, by 2015, the proportion of people without sustainable access to safe drinking water. By 2020, achieve a significant improvement in the lives of at least 100 million slum dwellers.

[10] Targets: Develop further an open, rule-based, predictable, nondiscriminatory trading and financial system. Address the Special Needs of the Least Developed Countries. Address the Special Needs of landlocked countries and small island developing states. Deal comprehensively with the debt problems of developing countries through national and international measures in order to make debt sustainable in the long term. In cooperation with developing countries, develop and implement strategies for decent and productive work for youth. In cooperation with pharmaceutical companies, provide access to affordable, essential drugs in developing countries. In cooperation with the private sector, make available the benefits of new technologies, especially information and communications. National specification of individual indicators and targets may differ from the ones reported in the WDI database.

Table 10. Ghana: Selected Medium-Term Indicators, 2006-11 (Baseline Scenario)

	2006 Proj.	2007 Proj.	2008 Proj.	2009 Proj.	2010 Proj.	2011 Proj.
	(Annual percentage change, unless otherwise specified)					
Real GDP	6.0	6.0	6.3	6.4	6.5	6.5
Real GDP per capita	3.4	3.3	3.6	3.8	3.9	3.8
GDP deflator	14.8	9.5	8.3	7.9	7.3	5.9
Consumer price index (annual average)	10.8	8.8	7.1	6.3	5.4	4.4
Exports, f.o.b.	31.5	5.9	3.0	3.9	2.9	1.4
Imports, f.o.b.	19.3	8.1	5.9	7.3	4.2	5.3
Export volume	12.1	2.6	4.9	4.8	4.6	3.9
Import volume	8.8	3.8	6.4	7.9	4.8	5.1
	(Percent of GDP, unless otherwise specified)					
Total revenue	22.3	22.7	22.1	21.7	21.5	21.4
Grants	6.3	5.2	4.0	3.6	3.7	3.7
Total expenditure	32.8	31.1	29.5	28.5	28.4	28.5
Capital expenditure	14.3	14.4	13.0	12.1	12.2	12.5
Real growth rate (using GDP deflator)	26.7	6.1	-4.1	-0.6	7.7	8.7
Overall balance (excluding grants)	-10.5	-8.4	-7.4	-6.8	-7.0	-7.1
Overall balance (including grants)	-4.6	-3.4	-3.5	-3.3	-3.4	-3.5
Domestic primary balance	-1.3	0.8	1.0	0.9	0.5	0.3
Net domestic borrowing	0.2	-0.1	0.2	0.0	0.2	0.6
Net foreign financing	4.4	3.5	3.3	3.3	3.2	2.9
Additional resources needed to finance investment plan	0.0	0.0	0.0	0.0	0.0	0.0
Domestic debt (net)	8.3	7.0	6.3	5.5	5.0	5.1
Current account balance (after grants)	-5.3	-4.7	-5.2	-6.2	-6.2	-6.2
NPV of external debt outstanding [1]	19.8	20.4	20.3	20.3	20.6	20.8
Percent of exports of goods and services	50.5	56.6	62.6	68.8	76.1	83.4
	(Millions of U.S. dollars, unless otherwise specified)					
Gross international reserves	2,090	2,427	2,868	3,239	3,677	4,082
Months of imports of goods and services	2.9	3.2	3.6	3.9	4.4	4.4

Sources: Data provided by Ghanaian authorities; and IMF staff estimates and projections.

[1] Based on the low-income country debt sustainability analysis framework.

Table 11. Ghana: Balance of Payments, 2006-11
(Millions of U.S. dollars; unless otherwise specified)

	2006	2007	2008	2009	2010	2011
			Proj.			
Exports f.o.b.	3,684	3,902	4,018	4,173	4,295	4,357
Cocoa beans and cocoa products	1,002	1,000	999	1,018	1,043	1,079
Gold	1,430	1,604	1,653	1,765	1,833	1,801
Timber and timber products	207	255	274	289	303	320
Others	1,046	1,043	1,091	1,102	1,116	1,158
Imports, f.o.b.	-6,377	-6,891	-7,298	-7,831	-8,163	-8,593
Non-oil	-4,803	-5,106	-5,232	-5,691	-5,939	-6,225
Oil	-1,575	-1,785	-2,066	-2,139	-2,224	-2,368
Trade balance	-2,693	-2,989	-3,280	-3,658	-3,868	-4,236
Services (net)	-564	-538	-480	-475	-527	-333
Of which : interest payments	-71	-83	-100	-115	-129	-142
Private transfers (net)	2,094	2,181	2,216	2,277	2,307	2,273
Current account balance (before grants)	-1,163	-1,346	-1,544	-1,856	-2,087	-2,296
Official transfers (net)	480	653	684	690	773	857
Current account balance (after grants)	-683	-693	-860	-1,166	-1,314	-1,439
Capital account	1,120	1,090	1,361	1,597	1,811	1,914
Official capital, including grants and loans (net)	763	513	555	620	681	658
Inflows	4,580	662	697	733	798	884
Amortization	-3,817	-149	-142	-113	-117	-227
Private capital	357	577	806	977	1,131	1,257
Other capital and errors and omissions	-34	-60	-60	-60	-60	-70
Of which: Errors and omissions	0	0	0	0	0	0
Overall balance	403	337	441	371	437	405
Financing	-403	-337	-441	-371	-437	-405
Net international reserves (- is increase)	-403	-337	-441	-371	-437	-405
Of which: use of Fund credit	-265	0	0	0	0	0
Disbursements (PRGF)	116	0	0	0	0	0
Repayments (PRGF)	-381	0	0	0	0	0
Exceptional financing	0	0	0	0	0	0
Memorandum items:						
Current account balance (in percent of GDP)						
Excluding official transfers	-9.0	-9.2	-9.3	-9.8	-9.9	-9.9
Including official transfers	-5.3	-4.7	-5.2	-6.2	-6.2	-6.2
Gross international reserves (US$ millions)	2,090	2,427	2,868	3,239	3,677	4,082
Months of imports of goods and services	2.9	3.2	3.6	3.9	4.4	4.5
Cocoa exports						
Volume (in thousands of tons)	630	590	590	602	614	626
Price (in US$ per ton)	1,550	1,650	1,640	1,630	1,630	1,651
External debt service						
Percent of exports of GNFS	11.7	4.3	4.4	4.1	4.3	6.3
Percent of government revenue	20.6	6.9	6.5	5.5	5.4	7.4
Terms of trade (percent change)	7.0	-0.9	-1.3	-0.3	-1.1	-2.5

Sources: Bank of Ghana; and IMF staff estimates and projections.

Table 12. Ghana: Selected Medium-Term Indicators, 2006-11 (Alternative Scenario A)

	2006 Proj.	2007 Proj.	2008 Proj.	2009 Proj.	2010 Proj.	2011 Proj.
	(Annual percentage change, unless otherwise specified)					
Real GDP	6.0	6.2	7.0	8.0	8.5	8.5
Real GDP per capita	3.4	3.5	4.4	5.3	5.8	5.8
GDP deflator	14.8	9.5	8.3	7.9	7.3	5.9
Consumer price index (annual average)	10.8	8.8	7.1	6.3	5.4	4.4
Exports, f.o.b.	31.5	12.0	9.2	9.7	10.2	10.3
Imports, f.o.b.	19.3	11.5	8.8	8.7	7.8	9.9
Export volume	12.1	7.1	6.9	6.3	6.5	8.2
Import volume	8.8	7.1	9.3	9.4	8.4	9.8
	(Percent of GDP, unless otherwise specified)					
Total revenue	22.3	23.1	22.7	22.3	22.3	22.3
Grants	6.3	5.2	3.9	3.5	3.5	3.5
Total expenditure	32.8	31.7	32.9	33.0	32.5	29.3
Capital expenditure	14.3	14.9	16.3	16.3	15.7	12.5
Real growth rate (using GDP deflator)	26.7	10.5	17.0	7.7	4.8	-13.6
Overall balance (excluding grants)	-10.5	-8.7	-10.2	-10.6	-10.1	-7.0
Overall balance (including grants)	-4.6	-3.6	-6.4	-7.2	-6.7	-3.7
Domestic primary balance	-1.3	0.6	-1.8	-2.8	-2.4	0.7
Net domestic borrowing	0.1	-0.3	-0.2	-1.3	-1.1	0.5
Net foreign financing	4.4	3.5	3.3	3.2	3.1	2.8
Additional resources needed to finance investment plan	0.0	0.5	3.3	4.2	3.6	0.4
Domestic debt (net)	8.3	6.8	5.6	4.6	4.0	3.9
Current account balance (after grants)	-5.3	-4.9	-5.3	-5.5	-5.5	-5.2
NPV of external debt outstanding [1]	19.8	20.5	24.2	28.3	31.3	30.2
Percent of exports of goods and services	50.5	54.7	69.0	86.0	99.9	99.6
	(Millions of U.S. dollars, unless otherwise specified)					
Gross international reserves (end of period)	2,090	2,395	2,811	3,277	3,813	4,407
Months of imports of goods and services	2.8	3.0	3.3	3.6	4.0	4.1

Sources: Data provided by Ghanaian authorities; and IMF staff estimates and projections.

[1] Based on the low-income country debt sustainability analysis framework.

Table 13. Ghana: Selected Medium-Term Indicators, 2006-11 (Alternative Scenario B)

	2006 Proj.	2007 Proj.	2008 Proj.	2009 Proj.	2010 Proj.	2011 Proj.
	(Annual percentage change, unless otherwise specified)					
Real GDP	6.0	6.0	6.3	6.4	6.5	6.5
Real GDP per capita	3.4	3.3	3.6	3.8	3.9	3.8
GDP deflator	14.8	9.5	8.3	7.9	7.3	5.9
Consumer price index (annual average)	10.8	8.8	7.1	6.3	5.4	4.4
Exports, f.o.b.	31.5	7.3	4.7	3.9	4.1	3.6
Imports, f.o.b.	19.3	11.5	8.5	6.4	4.6	4.7
Export volume	12.1	5.2	5.2	4.7	5.2	5.0
Import volume	8.8	7.1	9.1	7.0	5.2	4.5
Total revenue	22.3	23.1	22.8	22.3	22.1	22.0
Grants	6.3	5.2	4.0	3.6	3.7	3.8
Total expenditure	32.8	31.7	33.0	33.3	33.0	29.9
Capital expenditure	14.3	14.9	16.4	16.5	16.1	13.0
Real growth rate (using GDP deflator)	26.7	10.2	16.8	7.1	4.1	-14.1
Overall balance (excluding grants)	-10.5	-8.7	-10.3	-11.0	-10.9	-7.9
Overall balance (including grants)	-4.6	-3.6	-6.4	-7.5	-7.3	-4.3
Domestic primary balance	-1.3	0.6	-1.8	-3.0	-2.8	0.3
Net domestic borrowing	0.2	-0.4	-0.3	-0.1	0.3	0.9
Net foreign financing	4.4	3.5	3.3	3.3	3.2	3.0
Additional resources needed to finance investment	0.0	0.5	3.3	4.3	3.8	0.4
Domestic debt (net)	8.3	6.8	5.6	4.7	4.5	4.9
Current account balance (after grants)	-5.4	-6.1	-7.4	-7.9	-7.7	-7.0
NPV of external debt outstanding [1]	19.8	20.5	24.4	29.0	32.7	32.2
Percent of exports of goods and services	50.5	56.5	73.6	96.2	117.1	122.9
	(Millions of U.S. dollars, unless otherwise specified)					
Gross international reserves	2,090	2,223	2,301	2,376	2,527	2,795
Months of imports of goods and services	2.8	2.8	2.8	2.8	2.9	3.1

Sources: Data provided by Ghanaian authorities; and IMFstaff estimates and projections.

[1] Based on the low-income country debt sustainability analysis framework..

Table 14. Ghana: Revised Schedule of Disbursements Under the PRGF Arrangement, 2003-06 [1]

Schedule [2]			Updated Schedule			Conditions
Date	Amount in SDRs	Percent of quota	Date	Amount in SDRs	Percent of quota	
May 2003	26,350,000	7.14	May 2003	26,350,000	7.14	Board approval of the arrangement
December 2003	26,350,000	7.14	December 2003	26,350,000	7.14	Completion of first review
July 2004	26,350,000	7.14	July 2004	26,350,000	7.14	Completion of second review
July 2005	26,350,000	7.14	July 2005	26,350,000	7.14	Completion of third review
October 2005	26,350,000	7.14	June 2006	26,350,000	7.14	Completion of fourth review (end-June 2005 test date)
March 2006	26,350,000	7.14	June 2006	26,350,000	7.14	Completion of fifth review (end-December 2005 test date)
October 2006	26,400,000	7.15	October 2006	26,400,000	7.15	Completion of sixth (final) review (end-June 2006 test date)
Total	184,500,000	50.00		184,500,000	50.00	

[1] Ghana's quota is SDR 369 million.
[2] EBS/05/79.

APPENDIX I

Ghana: Relations with the Fund
(As of August 31, 2006)

I. Membership Status: Joined September 20, 1957; Article VIII.

II. General Resources Account	SDR Million	% Quota
Quota	369.00	100.0
Fund holdings of currency	369.00	100.0

III. SDR Department	SDR Million	% Allocation
Net cumulative allocation	62.98	100.0
Holdings	0.12	0.19

IV. Outstanding Purchases and Loans	SDR Million	% Quota
PRGF arrangements	79.05	21.42

V. Latest Financial Arrangements

Type	Approval Date	Expiration Date	Amount Approved (SDR Million)	Amount Drawn (SDR Million)
PRGF	05/09/2003	10/31/2006	184.50	158.10
PRGF	05/03/1999	11/30/2002	228.80	176.22
ESAF	06/30/1995	05/02/1999	164.40	137.00

VI. Projected Payments to Fund:
(SDR million; based on existing use of resources and present holdings of SDRs):

	Forthcoming				
	2006	2007	2008	2009	2010
Principal					
Charges/interest	0.81	2.85	2.86	2.85	2.85
Total	0.81	2.85	2.86	2.85	2.85

VII. Implementation of HIPC Initiative:

	Enhanced Framework
Commitment of HIPC assistance	
Decision point date	February 2002
Assistance committed by all creditors (US$ million) [1]	2,186.00
Of which: IMF assistance (US$ million)	112.10
(SDR equivalent in millions)	90.05
Completion point date	July 2004
Disbursements of IMF assistance (SDR million)	
Assistance disbursed to the member	90.05
Interim assistance	25.06
Completion point	64.99
Additional disbursement of interest income [2]	4.25
Total disbursements	94.30

[1] Assistance committed in the original framework is expressed in net present value (NPV) terms at the completion point; assistance committed in the enhanced framework is expressed in NPV terms at the decision point.

[2] In the enhanced framework, an additional disbursement is made at the completion point corresponding to interest income earned on the amount committed at the decision point but not disbursed during the interim period.

VIII. Implementation of MDRI Assistance:

I. Total debt relief (SDR million) [1]	265.39
Of which: MDRI	220.04
HIPC	45.35

II. Debt relief by facility (SDR million)

	Eligible Debt		
Delivery Date	**GRA**	**PRGF**	**Total**
January 2006	N/A	265.39	265.39

[1] The Multilateral Debt Relief Initiative (MDRI) provides 100 percent debt relief to eligible member countries that qualify. The relief covers all debt owed to the Fund as of December 31, 2004, that is outstanding at the time the member qualifies for relief. The MDRI is financed by bilateral contributions, the Fund’s own resources, and resources already disbursed to the member under the HIPC Initiative (see Section VII above).

Safeguard Assessment:

In conformity with Fund policy, the Bank of Ghana was subject to a safeguards assessment with respect to the PRGF arrangement approved on May 9, 2003. The assessment, which was completed on October 15, 2003, made specific recommendations to address vulnerabilities in the external audit, financial reporting, internal audit, and internal controls areas. The authorities confirmed implementation of all the measures proposed by staff.

Exchange Rate Arrangement:

On February 2, 1994, Ghana accepted obligations under Article VIII, Sections 2(*a*), 3, and 4, of the Fund's Articles of Agreement. The authorities describe their exchange rate regime as a managed float with no preannounced path, using the U.S. dollar as the intervention currency. However, the relative stability of the cedi-U.S. dollar exchange rate suggests that the de facto exchange rate regime is akin to a conventional fixed peg.[13] Ghana's de facto exchange rate regime has been reclassified accordingly, based on the stable behavior of the cedi vis-à-vis the U.S. dollar over the last 12 months. The system is free of restrictions on payments and transfers for current international transactions. At the end of September 2006 the average exchange rate for transactions in the interbank market was ¢9,210 per U.S. dollar.

Article IV Consultation and Current Fund Arrangement:

On May 9, 2003, the Executive Board concluded the 2003 Article IV Consultation with Ghana (*IMF Country Report 03/133*), endorsed the Ghana Poverty Reduction Strategy as a sound basis for promoting growth and reducing poverty, and approved a three-year arrangement under the PRGF amounting to SDR 184.5 million (50 percent of quota). On June 20, 2005, the Executive Board concluded the 2005 Article IV Consultation with Ghana (*IMF Country Report 05/292)* and the third review under the PRGF arrangement (www.imf.org/external/GHA/index.htm).

[13] According to the definition of IMF, Annual Report on Exchange Rate Arrangements and Exchange Restrictions (2005), the country (formally or de facto) pegs its currency at a fixed rate to another currency or a basket of currencies. The exchange rate may fluctuate within a narrow margin of less than 1 percent around a certain rate—or the maximum and minimum value of the exchange rate may remain within a narrow margin of 2 percent—for at least six months. The exchange rate may exceed the 2 percent range for at most five consecutive days during two instances within the six-month reference period, except for step devaluations and revaluations.

FSAP Participation:

Ghana participated in the FSAP in 2000–01, and a Financial System Stability Assessment (FSSA) was issued to the Executive Board in 2001 (SM/01/177; 6/19/01). An FSAP update was presented to the Board in December 2003.

Technical Assistance, 2002–June 2006:

Fiscal Affairs Department: *All Ministry of Finance:* peripatetic advisor on establishing a large-taxpayers' unit, February 2002–January 2003; review of public expenditure management reforms and assessment of long-term advisor, August 2002 and March 2003; tax peripatetic, May 2003; fiscal ROSC, February 2004; regional advisor on public expenditure management, August 2004–August 2005; evaluation of impact of new pricing mechanism for petroleum products on poor and vulnerable households, January 2005.
International Capital Markets: *Bank of Ghana*: pilot study of access to private capital, May 2003. *Bank of Ghana and Ministry of Finance*: access to international capital markets, November 2004.
Legal Department: *Bank of Ghana:* the remittance market, April–May, 2006.
Monetary and Financial Systems Department: *All Bank of Ghana (with Ministry of Finance)*: accounting and internal audit reform, July 2002, November 2002, and March 2003; foreign exchange market, government securities market, and banking system issues, April 2003; joint FSAP follow-up with the World Bank, June 2003; multitopic technical assistance initiation, November 2004; improving monetary operations, banking supervision, and payment systems, May 2005.
Statistics Department: *Bank of Ghana:* money and banking statistics, July 2002 and January–February 2004. *Ghana Statistical Service:* national accounts statistics, September and October 2001, August, November, and December 2002, and May and September 2003; national accounts and prices, March and October 2004, April–May 2005, April–May 2006, and September 2006; government finance statistics, March 2005. *Ministry of Finance and Economic Planning:* government finance statistics, May–June 2006.

Resident Representative:

The Fund has had a Resident Representative office in Accra since June 1985.
Mr. M. Arnold McIntyre, the current resident representative, assumed the post in August 2006.

APPENDIX II

Ghana: IMF-World Bank Relations

(As of July 14, 2006)

I. Partnership in Ghana's Development Strategy

1. Ghana's development objectives are stated in the Ghana Growth and Poverty Reduction Strategy (GPRS II) presented to parliament in October 2005. The objectives of the GPRS are to create an environment favorable to private sector–led growth and sustainable poverty reduction, and to create room within the Government's budget for increased expenditures on education, health, and other priority services. The overarching objective of GPRS II is for Ghana to attain middle-income status by 2015, which entails maintaining annual real GDP growth rates of at least 6 percent. Sustaining high real GDP growth rates are expected to reflect in turn positive social change and improvements in the quality of life for everyone.

2. The Bank and Fund teams are closely coordinating their policy advice to the Ghanaian authorities. There is collaboration in terms of common objectives and joint support for implementation of Ghana's GPRS, through, e.g., joint assessments such as the 2006 Joint Staff Advisory Note on the GPRS II and the 2005 Annual Progress Report, the 2004 HIPC Expenditure Tracking Assessment and Action Plan (AAP), the mid-2004 assessment of progress toward the completion point under the Enhanced HIPC Initiative, and the update of the findings of the 2000–01 Financial Sector Assessment Program (FSAP).

3. The GPRS II, and before that the Ghana Poverty Reduction Strategy (GPRS I), created the momentum for a group of development partners (DPs) to align their assistance through a multidonor budgetary support (MDBS) agreement with the Government of Ghana (GoG). The GoG and DPs consider the MDBS to be the basis for supporting implementation of GPRS II through the budget. Also, building on gains made over the last few years, GoG representatives and their development partners (nine bilaterals and four multilaterals, including the World Bank) agreed on February 25, 2005, to work together according to the principles established in "Harmonization and Alignment in Ghana for Aid Effectiveness: a Common Approach for Ghana and its Development Partners."

4. On June 21, 2006, Ghana held a Consultative Group (CG) meeting to articulate the targets of the GPRS II and outline the support it expected from DPs. The CG documentation included a prioritized result matrix, an aid harmonization and effectiveness action plan, and a support overview indicating actual and projected disbursements by DPs by sector and pillar of the GPRS II. The overview was designed to help align DP support with national strategies and the budget so that funding can be more predictable. In the future, CG meetings will be held every June to allow the government to better plan the budget process. Also, the partners plan to draft a Joint Assistance Strategy (JAS) for the GPRS II by late 2006. The Bank Group's next Country Assistance Strategy, which will emerge from the JAS, is expected to be presented to the Board early in 2007.

II. THE WORLD BANK GROUP COUNTRY ASSISTANCE STRATEGY AND PORTFOLIO

5. The Bank Board discussed the Country Assistance Strategy (CAS) for Ghana (FY04–07), prepared jointly with the International Finance Corporation (IFC), on March 16, 2004. The CAS was designed to support the GPRS by effectively helping Ghana achieve its development objectives; it complements the interventions of other donors by focusing on three areas: (i) sustainable growth and job creation, (ii) service provision for human development, and (iii) governance for empowerment. The Board discussed a CAS progress report (CAS PR) on June 15, 2006. The CAS PR confirms that the strategy laid out in the 2004 CAS can continue to guide the Bank's program for Ghana.

6. The Bank Group strategy is to help the government: (i) maintain the progress made in achieving macroeconomic stability in recent years while pursuing accelerated growth by improving the investment climate and harnessing sources of growth; (ii) build human capital by improving delivery of human development services, and (iii) promote good governance and public sector reforms. The CAS for Ghana was prepared after consultations with government and stakeholders, including meetings throughout the country with civil society, the private sector, and others. Another aspect of the Bank group's strategy in Ghana is to deepen its collaboration with other development partners through the MDBS framework, partnership programs like the Health SWAp, and other sector support, and to further harmonize approaches in areas like analytical work, assessment of the fiduciary underpinnings for sector and general budget support, and meeting and mission management.

7. Since its Board reviewed the CAS in March 2004, the Bank program has been important in financing Ghana's development program. Total disbursements rose to US$332 million in 2005, about 10 percent of total government expenditures. Actual disbursements for FY04–06 are expected to reach US$ 917 million (including the PRSC 4 presented at the Board on June 15, 2006), meaning that the Bank had delivered almost its entire planned CAS program by mid-2006.

8. The Bank's cumulative commitments to Ghana as of July 14, 2006, amounted to US$29 billion for 154 operations. The 24 active projects total US$1.35 billion, of which US$536 million is still undisbursed. The portfolio is diverse: It consists of a series of single-tranche Poverty Reduction Support Credits (PRSCs) for continued support of policies and reforms articulated in the GPRS II, complemented by programs in health and education, agriculture, energy, roads, community water, and other infrastructure in both rural and urban areas. Annex 1 summarizes World Bank operations in Ghana. The performance of the portfolio is generally satisfactory, though the implementation of some projects is delayed.

9. Given the emphasis on growth in the GPRS II, the Bank program for the remainder of the CAS period is expected to sharpen its focus on growth. The continued budget support through the PRSCs and new operations is designed to support government initiatives in information technology and telecommunications, the expansion of energy supply services, the extension of the road network, and the strengthening of natural resource management.

10. Nonlending services include updates of the core diagnostics, as well as targeted analyses to strengthen the analytical base for the assistance program supported by the

FY04–07 CAS. In FY04, this work consisted of a CFAA update; a country economic memorandum on public policy, growth, and poverty; a financial sector assessment update; an accounting and audit assessment; a report on policy options facing the new National Health Insurance Scheme; a public expenditure review (PER) carried out with other MDBS partners; and the first poverty and social impact analysis, which focused on electricity tariffs. In FY05, nonlending services included policy papers on the electricity and petroleum sectors, the joint WB-DFID-AFD study on natural resource management and economic growth, the external review of public financial management (ERPFM) prepared with other MDBS partners, and analytical work on policy options for the GPRS update and on monitoring and evaluation. In FY06, nonlending services so far have been an update of the external review of public financial management (ERPFM), also prepared with MDBS partners; a youth and employment study; and a country environmental assessment.

III. IMF-World Bank Collaboration in Specific Areas

11. The Bank and Fund teams are closely coordinating their policy advice to the Ghanaian authorities; they have common objectives and together they support Ghana's GPRS and determine structural conditionality.

12. **Areas where the Fund leads**. The IMF leads the policy dialogue on macroeconomic policies, such as fiscal and monetary policies. The IMF has supported Ghana's poverty reduction efforts through several arrangements under the Poverty Reduction and Growth Facility (PRGF). PRGF reforms center on measures to substantially raise revenue to make room for increased poverty-related spending and development needs, strengthen public expenditure management, and use monetary policy to achieve single-digit inflation. The fourth and fifth reviews were completed on June 12, 2006.

13. **Areas where the Bank leads**. The World Bank leads the policy dialogue on economic reforms in such sectors as infrastructure, which encompasses energy, roads, and telecommunications, water, and sanitation; urban and local government; agriculture and rural development; human development; and private sector development and public sector reform.

- **Infrastructure** accounts for about 56 percent of commitments. The Road Sector Development SIL has the objective of achieving sustainable and regionally equitable improvements in the supply and performance of road transport services. The Bank Group's engagement in energy has grown to include Economic Sector Work (ESW) and Poverty and Social Impact Analysis (PSIA) and investment lending to support the operational efficiency of the national utility companies. Meanwhile IFC is supporting private investment in power generation, and an IDA guarantee supports the country's participation in the West African Gas Pipeline project. IDA also plans to support Ghana's participation in the West Africa Power Pool project and an operation planned for FY07 to increase access to electricity. Dialogue on water and sanitation policy has been pivotal to major new investments in urban, small town, and community water access. In view of their importance for the delivery of reliable and cost-effective services, policy issues in the power and water sectors are also part of the PRSC-supported policy agenda. Based on new developments in the government's approach to telecommunications, the Bank Group is again engaged in dialogue on sector strategy, policy, and regulation and will support the country program through the proposed ICT/e-Ghana project.

- **Agriculture and rural development** accounts for about 12 percent of commitments; the intent is to find sustainable ways to increase agricultural productivity and diversification, deepen financial intermediation in rural areas, and rehabilitate land, forest, and wildlife resources. Beyond the cross-cutting issues tackled through the PRSCs, three IDA operations are supporting this agenda: the restructured Agriculture Services Sub-Sector Investment (AgSSIP) Project, the Rural Financial Services SIL, and the Community-Based Rural Development Project, which supports a comprehensive approach to rural economic growth and poverty reduction. There are also three Global Environment Facility (GEF) projects in support of the natural resource management agenda and preservation of biodiversity.

- In **human development,** five projects account for 17 percent of total lending. The Board approved financing for the Education Sector Project in February 2004. Twelve months earlier, it had approved the Second Health Sector Support Program prepared using a sector-wide approach (SWAp). In addition, Bank support addresses adult literacy, the prevention and treatment of HIV/AIDS, and new approaches to providing service to poor and marginalized groups. The PRSCs complement existing sector operations, leveraging their poverty focus by ensuring improved expenditure allocation (amount and structure) and addressing some of their financing implications for the poor (e.g., removal of school fees for girls in deprived areas and for disabled pupils across the country, and adopting a fee exemption policy for maternal deliveries). The PRSC-4 marks the transition from sector-targeted budget support through the health SWAP to general budget support directed to increasing government support to the health sector and sharpening the focus of government intervention on health outcomes.

- IDA's assistance in **private sector development** represents about 6 percent of Bank Group commitments. The Trade and Investment Gateway SIL responds to new government policies; it seeks to attract a critical mass of export-oriented investors to Ghana to accelerate export-led growth and facilitate trade. The Land Administration Project will help create a sustainable, fair, efficient, and decentralized land administration system to increase land tenure security. The PRSCs also support reforms tackling the high-cost business environment, facilitating trade, and encouraging financial intermediation. Further Bank Group support to private sector development is being provided by the new Micro, Small, and Medium Enterprise (MSME) Development Project, which promotes an enabling business environment and helps improve the capacity, competitiveness, and access to finance of small and medium enterprises. The joint donor Economic Management Capacity Building Project is designed to strengthen specific aspects of the financial sector and lay the foundation for public sector reforms.

14. **Areas of shared responsibility.** The IMF and World Bank staffs collaborate closely in supporting the GoG's structural reforms in budgeting, expenditure and financial management, public sector reform and privatization, and the financial sector, as shown

below. Bank support to governance and public sector management is provided mainly through the PRSC and through programs supporting public sector capacity building and decentralization.

15. **Budgeting, public expenditure management (PEM), and control.** The February 2004 joint Bank-Fund assessment of the government's PEM capacity confirmed encouraging progress since 2001, though there was still a need for continued improvement in managing the budget. That assessment, the FY04 PER, and the FY05 and FY06 External Reviews of Public Financial Management (ERPFM) identified priority PEM actions being supported by the PRSC/MDBS. Through the planned ERPFMs, dialogue is expected to continue to support the country's efforts to strengthen PFM performance. In the meantime, IMF technical assistance missions and a regional advisor are advising the government on budget formulation, monitoring of budget execution, and expenditure control.

16. **Public sector reform.** The Fund follows public service reforms through their impact on macroeconomic aggregates (wage bill, government expenditure) and discusses the macroeconomic tradeoffs the government faces in supporting a large public sector. Building on lessons learned, the government prepared a draft strategy and implementation plan for public sector reform that the cabinet approved last fall. The Bank's support for implementing the government strategy takes place in the context of the PRSCs and the Economic Management Capacity Building project, which supports results-oriented public sector reform.

17. **Public enterprise reform and divestiture.** The Fund monitors the financial position of large public enterprises in the energy and financial sectors because of their importance for public finances and macroeconomic stability. Bank assistance in this area is provided through sectoral projects. Bank dialogue in the energy sector, pursued through the Energy ESW and the PRSC, emphasizes the unbundling of the Volta River Authority, completion of the West Africa Gas Pipeline, and public-private partnership options for Electricity Company Ghana.

18. **Financial sector.** In July 2003 a joint Bank-IMF mission updated the 2000–01 FSAP. The FSAP Update confirmed that there had been some progress in the financial sector associated with improved macroeconomic stability, emphasized the diversity of financial institutions compared to the small size of Ghana's financial sector, and acknowledged the existence of strong international connections and local skills able to compete on equal terms with international institutions. Yet, the Update concluded, the financial sector is held back by weak financial intermediation compared to other sub-Saharan African countries. Problem areas are a shortage of long-term capital due to poor allocation of pension resources, the inefficiencies of the dominant Ghana Commercial Bank (GCB), and the need to further implement the legislative agenda. Several of the Update recommendations are being dealt with in the Government's comprehensive Financial Sector Strategic Plan (FINSSP), approved in 2003. The PRSC supports implementation of actions envisaged by the FINSSP, as does the Economic Management Capacity Building project.

APPENDIX II

ANNEX I

Financial Relations with the World Bank Group

(Active portfolio as of 14 July 2006, millions of U.S. dollars)

Credit/Grant Number	Fiscal Year	Project Name	IDA Credit	IDA Grant	GEF Grant	IDA Guar.[1]	Undis-bursed
26820-GH	1995	Ghana Thermal Power 5	175.6				11.7
20412-GH	1998	GEF Forest Biodiversity			8.7		1.3
31140-GH	1999	Trade Gateway and Investment	50.5				19.0
32460-GH	1999	National Functional Literacy	32.0				
33740-GH	2000	Rural Financial Services	5.1				0.3
34050-GH	2001	Agricultural Services	67.0				9.0
35540-GH	2002	Road Sector Development	220.0				36.5
50723-GH	2002	GEF Northern Savanna			7.6		3.9
37310-GH	2003	Health Sector Program II	57.6	32.4			
37430-GH	2003	Promoting Partnerships with Traditional Authorities		5.0			0.3
38170-GH	2004	Land Administration	20.5				13.6
38650-GH	2004	Education Sector Project	78.0				60.9
3890-GH	2004	Urban Environment and Sanitation	62.0				58.5
39620-GH	2005	Urban Water		103.0			78.2
39710-GH	2005	Small Towns Water Supply	26.0				21.8
39640-GH	2005	Community-Based Rural Development	60.0				44.9
33917-GH	2006	Multisectoral HIV/AIDS program	20.0				18.75
33719-GH	2006	Economic Management Capacity Building	25.0				24.25
31985-GH		Micro, Small, and Medium Enterprise Development	45.0				45.0
Total (number of credits/grants: 18)[2]			944.3	140.4	16.3		517.0
N.A.	2005	West Africa Gas Pipeline				8.0	
Total (number of partial risk guarantees: 1)						8.0	
N.A.	2005	West Africa Power Pool APL-1 Phase 1				40.0	

Source: World Bank

[1] IDA partial risk guarantee up to US$50.0 million (of which 25 percent is IDA commitment) to protect commercial parties against sovereign risk.

[2] Cumulative commitments for active projects total US$1.04 billion.

For additional information, please contact Philip Jespersen, Program Officer ((202) 477-73102), or Carlos Cavalcanti, Lead Economist, ((202) 458-89076)

APPENDIX III

Ghana: Recent Trends in Remittance Inflows[14]

Magnitude of remittances and their sources

- While accurate data on remittances in Ghana are not available, it is estimated that remittance inflows are the fastest-growing source of foreign exchange. The estimated increase in their share from 4 percent of GDP in 1995 to 15 percent in 2005 (and to a projected 17 percent in 2006) is a result of better recording as well as a stronger financial sector and a better investment climate in Ghana.

- Since 2004 remittances have become Ghana's single largest source of foreign exchange, exceeding receipts from both cocoa and gold exports. Remittances have also been a substantially higher and more stable source of financing than private capital inflows and official foreign aid.

- Most of the inflows (over 90 percent) originate in the United States, Canada, the United Kingdom, and the rest of the European Union, which are the main destinations of Ghana's migrants. Remittances are usually sent frequently (at least monthly) by close relatives who have been abroad for at least one year. Transfers, which are most often into urban areas, typically arrive within a day and are mostly collected in cash.

- The Bank of Ghana (BoG) has been improving its remittance statistics. In 2004 it introduced a new format for banks that provides a breakdown by receiving institution and source of transfer.

- The official data still do not capture informal remittances (brought by relatives or sent through informal transfer agents); the BoG believes these to be substantial.

Bringing remittance inflows into formal channels

- Banks are by far the main channel for receipt of remittances. The four major banks account for 60 percent or more sent through formal channels to Ghana; the rest of the banking system receives about 30-35 percent; and nonbank financial institutions get only 5-10 percent.

- Measures now being implemented to update financial services (such as introducing smart cards), modernize the payment system, and fortify supervision are likely to increase competition and improve the efficiency of the formal financial system.

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[14] This Appendix is based on the following documents: E.K.Y. Addison, 2004, "The Macroeconomic Impact of Remittances in Ghana"; R. Adams, 2006, "Remittances and Poverty in Ghana," WPS 3838; A. Higazi, 2005, "Informal Remittance Systems in Africa, Caribbean and Pacific: Ghana Country Study," R02CS008; USAID (2005), "Private Remittance Flows to Ghana Project Country Report (draft, October 2005).

- To give rural populations more access to the formal financial sector, an effort is being made to link bank branches in rural areas with urban banks. As part of this effort, the Millennium Challenge Corporation is supporting computerization of over 100 rural banks and connecting them with the rest of the banking system. This effort would also support the system for formal remittances.

The main macroeconomic impact

- In Ghana, remittances have helped lower the current account deficit, somewhat reducing the need for external borrowing and the risk of debt distress. According to Adams (2006),[15] remittances have probably reduced poverty by (i) increasing incomes (by more than 20 percent for the poorest 20 percent of households) and (ii) diversifying income. It appears that, on the margin, remittances have significantly increased (by 20-30 percent) investment in education and health.

[15] R. Admas, 2006, "Remittances and Poverty in Ghana," WPS 3838.

APPENDIX IV

October 18, 2006

Mr. Rodrigo de Rato
Managing Director
International Monetary Fund
700 19th Street, NW
Washington, DC 20431
USA

Dear Mr. de Rato:

1. Further to our letter of May 25, 2006, reviewing performance under our economic and financial program over the past few years and presenting our program for 2006, we would like to inform you that performance has continued to be satisfactory. Real GDP growth has continued to be strong, and inflation is well contained—apart from the impact of domestic fuel price increases (considering rising world prices and our recent adoption of a full-cost pass-through regime). Also, Ghana's external position has strengthened, particularly after the MDRI debt relief.

2. All but one of the quantitative and structural performance criteria for June 2006 were met (Tables 1 and 2). The ceiling on the net domestic assets of the central bank for June 2006 was exceeded because an adjuster for the transfer and use of the MDRI resources was inadvertently omitted from the Technical Memorandum of Understanding (TMU) attached to our letter of May 25, 2006. Also, after the IMF Executive Board approved our program for 2006, we decided to accelerate the transfer and use of MDRI resources in 2006 from the US$125 million indicated in our letter to US$200 million.

3. Furthermore, the omission of the adjuster also affected the performance criterion for net international reserves for June 2006, but it had no implications for the observance of the target because net international reserves overperformed with or without correction for the errors.

4. As anticipated in our letter of May 25, 2006, parliament approved a supplementary budget for 2006 in July. The supplementary budget concentrated on the appropriate use of the additional resources made available from the MDRI to achieve our development objectives—primarily by augmenting public investment in poverty sectors. Initially, we had thought of directing nearly all the additional resources to poverty-reducing activities, which would have further raised poverty-related expenditures from 8.5 percent of GDP in 2005 to 10.0 percent in 2006. After consultation with the general public and its representatives, we have dedicated a significant portion of the resources to rehabilitation of critical infrastructure, particularly in the urban and rural poor areas, to enhance the prospect of sustainable growth. This allocation is consistent with the poverty reduction strategy we forwarded to you earlier this year. Thus, the rise in poverty-related expenditures was limited to 9.3 percent of GDP in

2006. In essence, the total MDRI resources of US$229 million (1.8 percent of GDP) is being used about equally for poverty-reducing activities and growth-enhancing investment.

5. In the circumstances, we are requesting a waiver for the nonobservance of the quantitative performance criterion on the net domestic assets of the Bank of Ghana for end-June 2006. Moreover, for the remainder of our economic program for 2006, we have adjusted (a) the end-September and end-December 2006 indicative targets for net domestic assets and the net international reserves of the Bank of Ghana, and (b) the end-December 2006 indicative target for poverty-related expenditures to reflect the higher MDRI spending and to correct the omission of the adjuster.

6. We continue to believe that the policies specified in our letter of May 25, 2006, provide a strong basis for sustaining growth, reducing inflation, and alleviating poverty. However, we are prepared to take any additional measures necessary to achieve the economic and financial objectives of the 2006 program. The government will continue to provide the Fund with the information necessary for assessing progress in implementing our program as specified in the TMU, and we will consult with the Fund staff and management on any measures that may be appropriate at the initiative of the government or whenever the Fund requests a consultation. The government intends to make public both this letter and the staff report for the sixth and final review under the PRGF arrangement. Accordingly, the government of Ghana authorizes the IMF to arrange for these documents to be posted on the Fund's website once the Executive Board has completed its review.

7. With the above, we hereby request completion of the sixth and final review and disbursement of the seventh loan installment of SDR26.4 million.

Yours sincerely,

s/ s/

Hon. Kwadwo Baah-Wiredu, MP
Minister of Finance and Economic Planning

Dr. Paul A. Acquah
Governor, Bank of Ghana

Attachments (2)

Table 1. Ghana: Quantitative Performance Criteria and Indicative Targets, PRGF Arrangement, 2006 [1]
(Cumulative flows from beginning of calendar year 2006 to end of month indicated, unless otherwise indicated)

	End-March	End-June			End-September	End-December
	Prel.	Perf. criterion	Prel.	Status	Indicative	Indicative
			(Billions of cedis)			
Performance criteria						
Net domestic financing of government (ceiling)	1,722	2170	443	Met	870	215
Net domestic assets of the Bank of Ghana (ceiling) [2]						
Approved by Board	-3,682	-4,376	-4,063	Not met	-5,086	-1,939
Proposed [3]	...	...	...	...	-3,315	-137
Stock of net domestic banking sector credit to TOR (ceiling)	1,031	950	697	Met	300	200
			(Millions of U.S. dollars)			
Net international reserves of the Bank of Ghana (floor)						
Approved by Board	306	165	357	Met	225	160
Proposed [4]	...	...	...	...	316	403
The contracting or guaranteeing of new nonconcessional external debt with original maturity greater than or equal to 1 year by the government or the Bank of Ghana (ceiling) [5]	0	0	0	Met	0	0
Outstanding stock of external debt with original maturity of less than one year owed or guaranteed by the government or the Bank of Ghana (ceiling) [6]	50	100	0	Met	100	100
Stock of external payment arrears (ceiling) [7]	0	0	0	Met	0	0
			(Billions of cedis)			
Wage bill of the central government (ceiling)	2,148	5,387	4,987	Met	7,588	9,990
Indicative targets						
Reserve money stock (ceiling)	9,308	9,600	9,534	Met	9,720	11,900
Poverty related expenditures (floor)						
Approved by Board	1,800	4,242	4,613	Met	6,680	11,187
Proposed	...	...	...	...	...	11,024

Sources: EBS/06/72; Table 1a; Ghanaian authorities; and staff estimates.

[1] Definitions of line items and terminology are elaborated in the Technical Memorandum of Understanding (TMU). Variables are measured at end of month values, unless otherwise indicated.

[2] Based on the official end-March exchange rate 9,138.8 cedis/U.S. dollar.

[3] Assumes that US$200 million was transferred to and used by the government under the MDRI resources from the Fund.

[4] Assumes that US$200 million from the MDRI resources from the Fund was spent and absorbed.

[5] This is a continuous performance criterion. It applies not only to debt as defined in point No. 9 of the Guidelines on Performance Criteria with Respect to Foreign Debt adopted by Decision 12274-(00/123) of August 24, 2000 but also to commitments or contracted for which value has not been received, as specified in paragraph 20 of the TMU.

[6] The term "debt" has the meaning set forth in point No. 9 of the Guidelines on Performance Criteria with Respect to Foreign Debt adopted by Decision 12274-(00/123) of August 24, 2000, as specified in paragraph 19 of the TMU.

[7] This is a continuous performance criterion. The TMU stipulates the precise program definition of payment arrears.

Table 2. Ghana: Structural Performance Criterion and Benchmark for 2006

Condition	Status
Performance criterion	
• **June 30, 2006.** Present a report on the comprehensive review of the wage and salary structure of public service workers, with the aim of rationalizing the emolument structure, monetizing benefits, standardizing job titles, and updating job classifications.	**Completed**
Benchmark	
• **June 30, 2006.** The Ministry of Finance and Economic Planning will submit the Credit Reporting Bill to Cabinet for final approval.	**Completed**

Sources: EBS/06/72, and information provided by the Ghanaian authorities.

APPENDIX V

Ghana: Illustrative External Debt Sustainability Scenarios[16]

Ghana's risk of debt distress is moderate but depends significantly on the productivity of new investment. In the baseline scenario, which assumes continuation of current investment levels, all the external debt burden indicators are below their policy-dependent thresholds. These results confirm that following HIPC and MDRI, Ghana's capacity to take on new loans has increased, creating room for limited borrowing on nonconcessional terms for high return projects. Ghana's risk of debt distress would still be moderate in the authorities' scenario that assumes scaled up investment, which is projected to accelerate growth but will be partially financed by some nonconcessional borrowing (Alternative Scenario A). In this scenario, all the debt indicators also remain below their policy-relevant thresholds, but some of these thresholds are breached under the stress tests, pointing to higher risk of debt distress than in the baseline scenario. The alternative scenario B of nonconcessional borrowing with low growth illustrates the risk that if nonconcessional borrowing undertaken (as in the Alternative Scenario A) does not lead to high growth, Ghana will be at high risk of debt distress. In that case, the NPV of debt-to-exports ratio would exceed its policy-dependent threshold, and all the indicators would breach their thresholds in stress tests. The high risk of debt distress would be exacerbated in case of negative external shocks, for example, if remittances or official grants were to stagnate in the long term.

Background

Between 1993 and 2002, real GDP growth in Ghana averaged 4.4 percent a year, but rose appreciably to 5.2 percent in 2003, 5.6 percent in 2004 and 5.9 percent in 2005. The main sources of growth in the recent period have been the traditional sectors of agriculture, mining, construction, and trade but other sectors such as horticulture, tourism, and transport, communication and financial services are emerging as important contributors. The historical annual average real GDP growth rate for the past ten years used in this exercise is 4.7 percent.

Ghana has relied on external financing on concessional terms to facilitate its development. At the end of 2005, 88 percent of Ghana's external debt was owed to multilateral creditors (on concessional terms), 8 percent to bilateral creditors (after the HIPC completion point in July 2004), and 4 percent to commercial creditors. Since the recent

[16] These illustrative debt sustainability scenarios were conducted by the Fund staff and the Ghanaian authorities using the debt sustainability framework for low-income countries. The mission was not able to collect full and reliable data needed for inclusion of domestic debt (particularly contingent liabilities associated with the operations of the SOEs). Nevertheless, given the relatively low levels of domestic debt (8 percent of GDP) preliminary staff estimates suggest that inclusion would not significantly alter the results.

MDRI relief, the share of multilateral debt is projected to decrease to about 70 percent of total debt in 2006.

Ghana's external debt has shown marked improvement, from 131 percent of nominal GDP in 2000 to 22 percent in mid-2006. Debt was significantly reduced in 2004 as a result of the HIPC completion point debt cancellation (US$1,625 million) and again in 2006 with the MDRI relief from the IMF (US$381 million), the IDA (US$2,892 million) and the African Development Fund (US$486 million). In addition to debt relief, other factors helping to reduce the debt–to-GDP ratio were real GDP growth and, in recent years, large net current private transfers (remittances).

Baseline Scenario—Continuation of Investment Levels

The baseline scenario, developed on a conservative set of assumptions about economic policies and outcomes specified in Box 1, shows that **Ghana would be at moderate risk of debt distress.** The 2005 World Bank's Country Policy and Institutional Assessment (CPIA) ranks Ghana's policies and institutions as corresponding to those of a "strong performer." [17] The debt-burden thresholds for countries in this category are NPV of debt-to-exports of goods and services of 200 percent, NPV of debt-to-GDP of 50 percent, and debt service-to-exports of goods and services of 25 percent.

All of Ghana's external debt burden indicators remain below the policy-dependant thresholds during the projection period under the baseline scenario (Figure 1 and Tables 1 and 2). The NPV of debt-to-GDP ratio amounts to 20 percent in 2006, well below the threshold of 50 percent, and is projected to gradually increase to 30 percent by 2025. Debt service payments continue to be very manageable, reflecting recent HIPC and MDRI, as well as the fact that all Ghana's external public and publicly guaranteed debt is projected to be contracted on concessional terms. The results remain broadly the same with a stress of a somewhat lower growth rate to reflect potential negative impact of the current energy shortage, but with smaller margins from the thresholds. However, the NPV of debt-to-exports ratio increases significantly, but remains below its threshold over the years, underscoring the importance of implementing the relevant structural reforms to reduce vulnerability to external shocks.[18]

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[17] Ghana's CPIA ranking improved from 3.7 in 2004 to 3.9 in 2005; 3.75 is the cut-off between "medium" and "strong performer"

[18] Ghana was ranked among the top ten reformers of business environment in 2005/2006, according to the World Bank in *Doing Business 2007*.

Box 1. Macroeconomic Assumptions for Baseline Scenario

Real GDP grows at about 6.3 percent until 2011 and at 5.5 percent between 2012 and 2025. The continuation of current policies includes no nonconcessional borrowing and continued limited investment in key infrastructure. Over the long term, private consumption would be an important source of growth, due to increased income and continued inflows of remittances.

Exports of goods and services are conservatively projected to grow (in US$ terms) at an annual rate of 7.4 percent between 2006 and 2011, and 7.7 percent between 2012 and 2025, well below its historical average of 10 percent. In the long run, exports would be driven by nontraditional exports, which would grow at an annual rate of about 18 percent from 2012 onward and 11 percent between 2006 and 2011, albeit from a relatively small base.

Mainly, reflecting conservative export forecast **the external current account deficit** (excluding official transfers) is projected to increase from 9.1 to 9.9 percent of GDP a year during 2006-2011, and then gradually decrease to slightly below 7 percent in 2025. The **international reserve coverage** of imports is projected to stay at around 4 months of imports of goods and services throughout the projection period.

Foreign direct investment is envisaged to average only 3 percent of GDP a year between 2007 and 2011, and less than 2 percent a year between 2012 and 2025. **Official external loan financing on concessional terms** is projected to average about 4 percent of GDP between 2007 and 2011 and 3.5 percent thereafter.

The historical scenario (using in particular the historical growth rate of 4.7 percent a year) in this case shows that NPV of debt-to-GDP and NPV of debt-to-exports and debt service-to exports ratio would only approach their thresholds in the outer years. In the most extreme stress test, consisting of the combined shock to GDP, exports and foreign direct investment, the NPV of debt-to-GDP ratio would rise from 20 percent in 2006 to 46 percent in 2008 and decline gradually to 43 percent toward the end of the period (Figure 1a). Also, transfers are lower than projected, both the NPV of debt-to-export ratio (Figure 1b) and the debt service-to-exports (Figure 1c) would stay below their thresholds throughout the projection period.

Alternative Scenario A—Nonconcessional Borrowing with High Growth

The scenario of nonconcessional borrowing with high growth scenario (alternative A) is built on assumptions about economic policies and outcomes specified in Box 2 including nonconcessional borrowing to finance high return projects that would contribute to the acceleration of the growth rate. The results (Table 3 and Figure 2) show that Ghana would have only a moderate risk of debt distress also under this scenario.

Box 2. Macroeconomic and Borrowing Assumptions for Alternative Scenario A

Real GDP grows at about 7.4 percent until 2011 and at 6.3 percent between 2012 and 2025. Over the medium term growth is driven by large-scale investment (particularly in basic infrastructure, energy, etc.) and exports. Long term, private consumption (thanks to increased income and continued inflows of remittances) is an important source of growth.

Exports of goods and services are projected to grow (in US$ terms) by about 12 percent until 2011, driven by exports of nontraditional commodities and aluminum. Between 2012 and 2025, exports continue to grow by 7.6 percent. The authorities have identified sectors other than cocoa and gold as key for increasing diversification, and investment in infrastructure and structural reforms are expected to support this strategy.

The external current account deficit (excluding official transfers) in terms of GDP is below its ten-year average of 10.2 percent by about 1 percentage point of GDP between 2006 and 2011, also because remittances increase rapidly from low levels. The current account deficit, excluding official transfers, is projected to gradually decrease further to 6.2 percent of GDP, while the **international reserves coverage** of imports is projected to rise gradually to about 5 months by 2025.

Official external loan financing on concessional terms is projected to average about US$700 million a year—4 percent of GDP—between 2007 and 2011, and then fall to an average of 3.5 percent thereafter.

This scenario assumes that **borrowing on nonconcessional terms** will average 3 percent of GDP (US$470 million) a year between 2007 and 2011 and decline to 1 of GDP percent between 2012 and 2025. Based on **the market terms of financing** obtained by comparable countries, it is assumed that with a credit rating of B+, Ghana would likely get annual interest of Libor plus 4-4.5 percentage points on international markets for an annual interest rate of 9.5 –10 percent.

In this scenario, all of Ghana's external debt burden indicators remain below the policy-dependent thresholds throughout the projection period (Figure 2 and Table 3). The NPV of debt-to-GDP ratio amounts to 20 percent in 2006, well below the threshold of 50 percent, and then is assumed to increase until 2010 because of accelerated borrowing for public investment projects and no additional multilateral debt relief. This ratio is projected to remain roughly constant during 2011-25. While debt service increases over time because there is commercial borrowing, it should continue to be manageable as the majority of Ghana's external public and publicly guaranteed debt would still be contracted on concessional terms. Although it remains below its threshold over time, the NPV of debt-to-exports of goods and services ratio increases significantly; that is why it will be important for Ghana to continue implementing structural reforms to diversify the economy and accelerate exports.

Because the historical scenario is based on averages from 1996 to 2005, it reflects a period of macroeconomic destabilization (currency depreciation and low export and import growth). In this case, all debt burden ratios exceed the threshold indicators. While both internal economic conditions and the external environment have changed, this scenario illustrates the importance of preserving macroeconomic stability and further firming up debt management while accommodating some borrowing on commercial terms and aiming for higher growth.

Adverse macroeconomic shocks would have a severe negative impact on Ghana's external position. If there is a combined shock (to real GDP growth, exports growth, foreign direct investment (FDI), etc.), the ratio of Ghana's NPV of debt-to-GDP ratio would rise to 59 percent in 2009—above the policy-dependent threshold of 50 percent for half of the projected period. A similar shock would also significantly worsen the NPV of debt-to-exports ratio, which would reach 170 percent in 2009, much closer to the threshold of 200 percent.

Alternate Scenario B—Nonconcessional Borrowing with Low Growth

This scenario was based on the same GDP growth assumptions as in the baseline scenario (Box 1) and the nonconcessional borrowing assumptions of the alternative scenario A (Box 2). In this scenario, **Ghana would be at high risk of debt distress:** the NPV of debt-to-exports ratio would breach its threshold, and the debt service-to-exports ratio would approach the threshold in the outer years (Figure 3 and Table 4). Moreover, all stress tests and historical scenarios are at or above the thresholds.

Figure 3 shows worse results than Figures 1 and 2 because this scenario combines nonconcessional borrowing with generally weaker economic performance. In the most extreme stress tests—the combined shock of (i) lower real GDP growth, exports, and FDI, and (ii) less concessional borrowing—all policy-relevant thresholds are breached, including that for the debt service-to-exports ratio. Moreover, thresholds for the debt-to-GDP and debt-to-exports ratios are exceeded by wide margins.

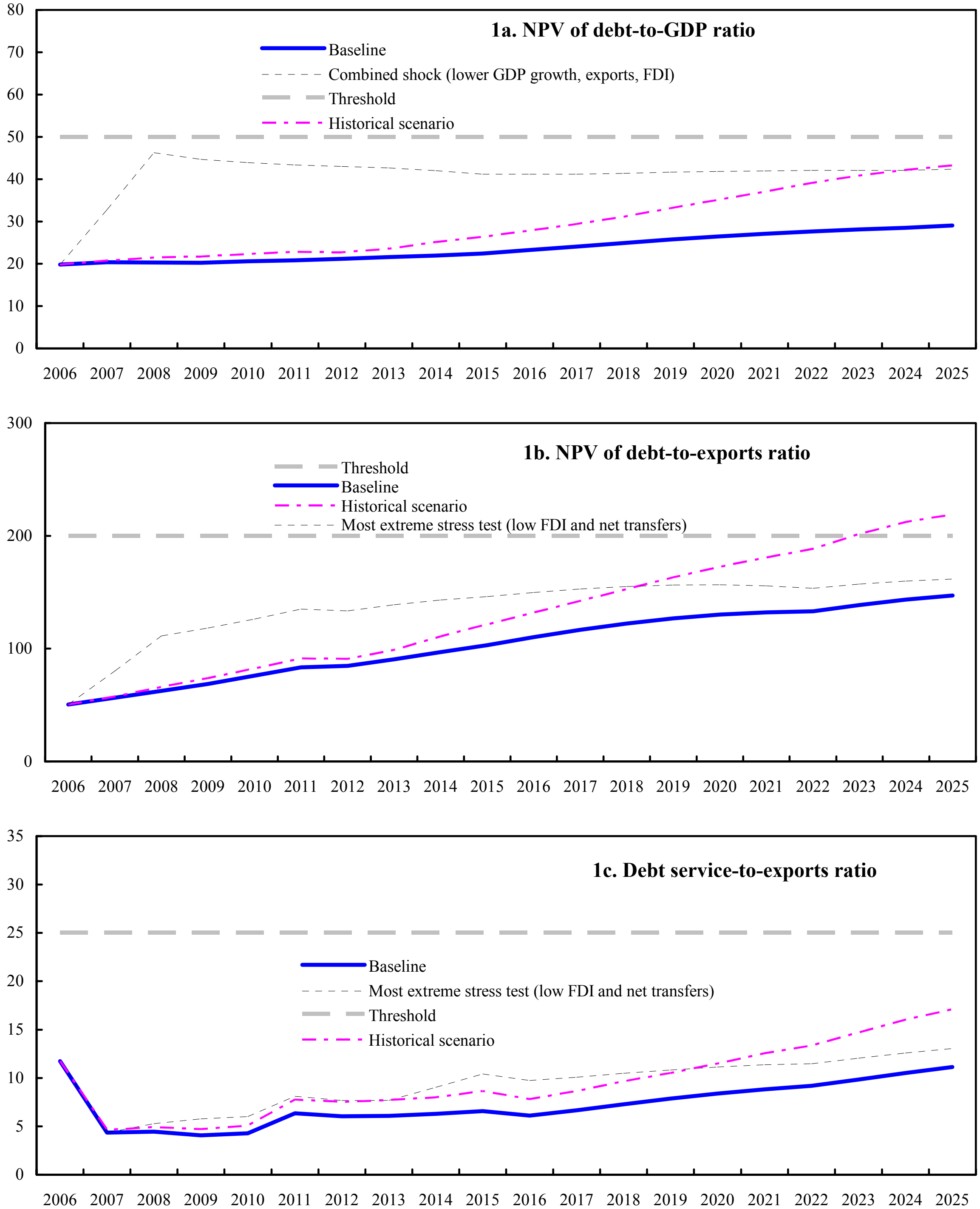
Discussions with the authorities

The authorities were highly aware of the risks associated with nonconcessional borrowing and the need to maintain debt sustainability. Better debt management continues to be a priority for them. Recently, the Ministry of Finance and Planning produced its own debt sustainability analysis (using the HIPC framework). The report also covers the evolution of debt aggregates and the government's strategy for managing domestic as well as external debt. For the coming years, the authorities are considering the possibility of cautiously contracting nonconcessional external debt if concessional aid remains below what they need to undertake critical investment projects. They thought that prudently accessing international markets would be essential to keep debt sustainable and send positive signals to foreign investors. They intend to draw on lessons from other countries.

Conclusions

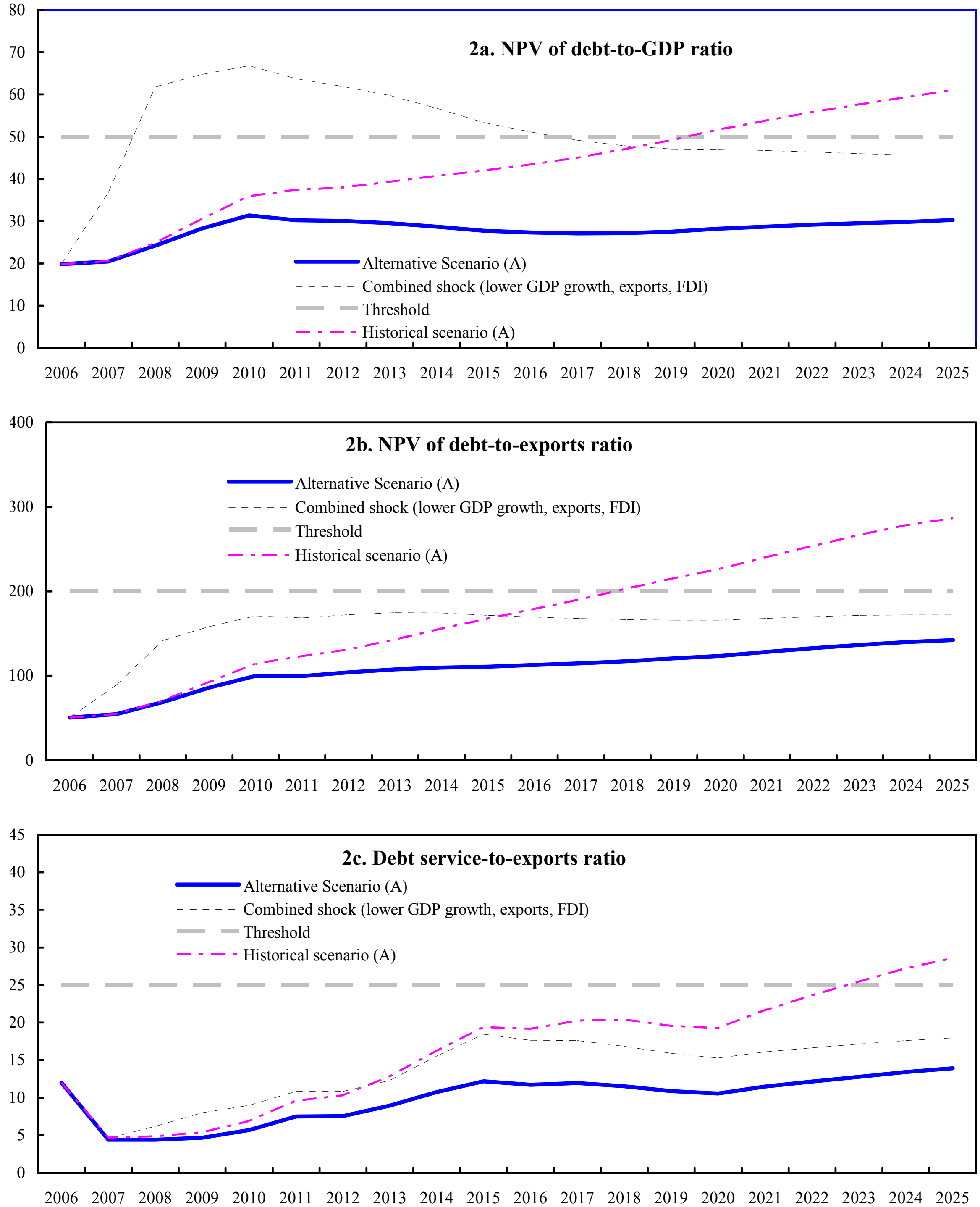
Ghana's risk of debt distress is moderate under the baseline scenario and under the authorities' Alternative Scenario A with nonconcessional borrowing and high growth. However, if nonconcessional borrowing were combined with low growth (Alternative Scenario B) debt sustainability would deteriorate substantially, particularly in case of negative macroeconomic shocks. To prevent that, Ghana should (i) exhaust all avenues for concessional aid; (ii) examine critically the productivity of investment projects to ensure that they will contribute to economic growth before undertaking them; and (iii) ascertain that any borrowing, concessional or nonconcessional, would not compromise debt sustainability. Keeping debt sustainable also requires that Ghana further build its management capacity to monitor debt and design and analyze loan proposals.

Figure 1. Ghana: Indicators of Public and Publicly Guaranteed External Debt Baseline Scenario, 2006-2025 (In percent)



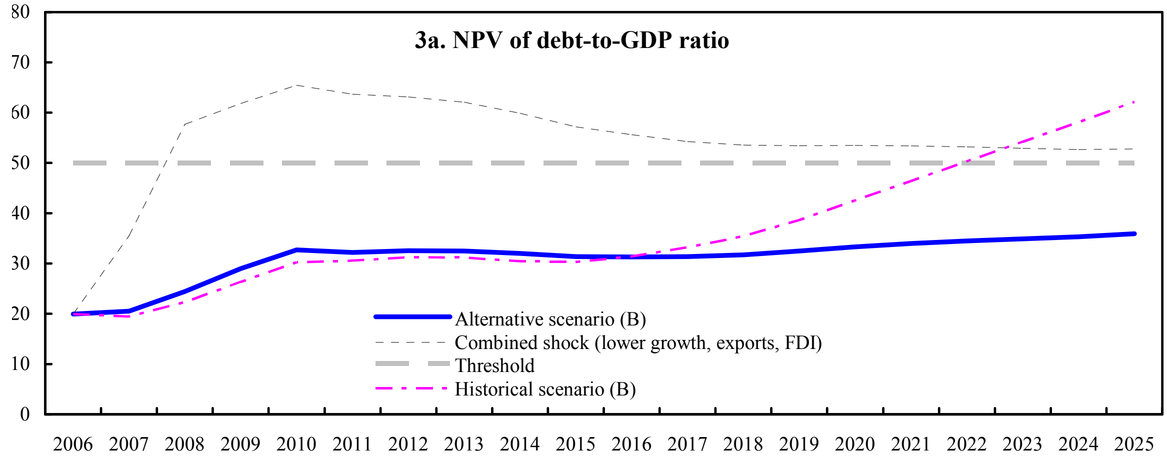
Sources: Staff projections and simulations.

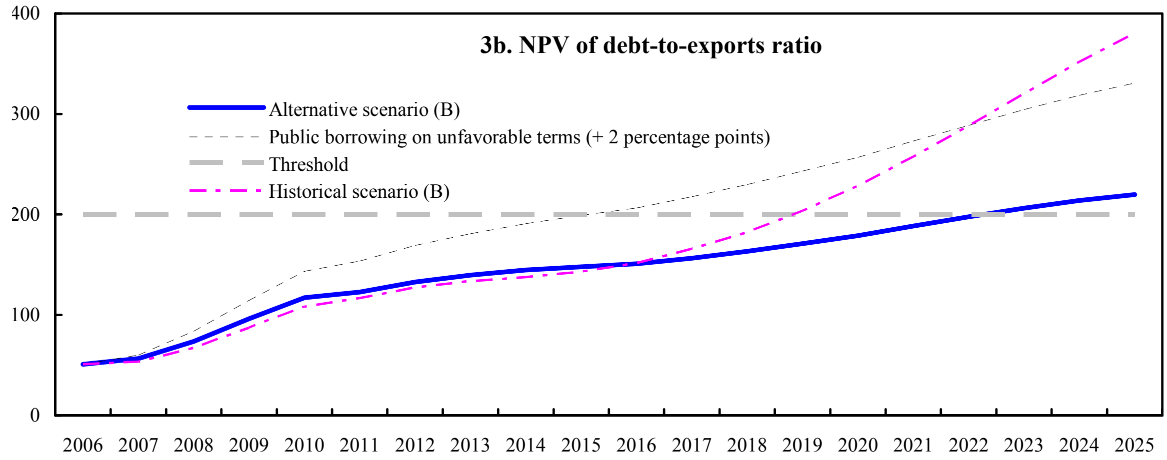
**Figure 2. Ghana: Indicators of Public and Publicly Guaranteed External Debt
Alternative Scenario A, 2006-2025
(In percent)**

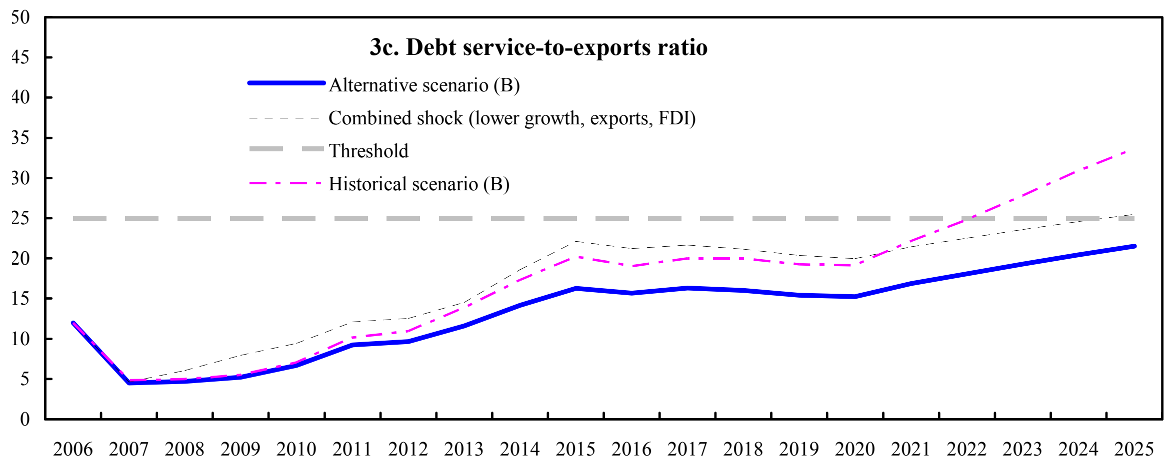


Sources: Staff projections and simulations.

Figure 3. Ghana: Indicators of Public and Publicly Guaranteed External Debt Alternative Scenario B, 2006-2025
(In percent)







Sources: Staff projections and simulations.

Table 1. Ghana: External Debt Sustainability Framework, Baseline Scenario, 2003-2025 1/
(In percent of GDP, unless otherwise indicated)

		Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	#	2003	2004	2005			2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2025	2012-25 Average
External debt (nominal) 1/		**99.0**	**72.7**	**59.2**			**21.2**	**22.1**	**22.9**	**23.5**	**24.3**	**24.9**		**28.2**	**33.9**	
o/w public and publicly guaranteed (PPG)		99.0	72.7	59.2			21.2	22.1	22.9	23.5	24.3	24.9		28.2	33.9	
Change in external debt		-0.5	-26.3	-13.5			-38.0	0.9	0.8	0.6	0.8	0.7		0.9	0.5	
Identified net debt-creating flows		-21.8	-15.0	-6.8			1.1	2.9	2.0	2.6	2.2	1.7		1.5	1.5	
Non-interest current account deficit		**-3.2**	**1.7**	**6.3**	**4.1**	**5.2**	**4.8**	**4.2**	**4.6**	**5.6**	**5.6**	**5.6**		**4.7**	**3.7**	4.1
Deficit in balance of goods and services		12.0	21.1	25.6			25.0	23.3	21.9	21.2	20.2	18.8		14.9	10.0	
Exports		40.7	39.3	36.1			39.2	36.0	32.5	29.4	27.1	25.0		21.1	19.7	
Imports		52.7	60.4	61.7			64.2	59.3	54.4	50.7	47.2	43.8		36.0	29.8	
Net current transfers (negative = inflow)		-15.7	-20.6	-19.7	-13.3	4.9	-19.9	-19.3	-17.4	-15.7	-14.6	-13.5		-10.4	-6.1	-9.2
Other current account flows (negative = net inflow)		0.4	1.3	0.5			-0.3	0.2	0.1	0.1	0.1	0.3		0.1	-0.2	
Net FDI (negative = inflow)		**-1.0**	**-3.7**	**-1.4**	**-1.5**	**0.9**	**-1.3**	**-0.7**	**-1.9**	**-2.3**	**-2.7**	**-3.1**		**-2.3**	**-1.6**	-2.1
Endogenous debt dynamics 2/		**-17.6**	**-13.0**	**-11.8**			**-2.4**	**-0.6**	**-0.6**	**-0.7**	**-0.8**	**-0.8**		**-0.8**	**-0.6**	
Contribution from nominal interest rate		1.5	0.9	0.7			0.5	0.5	0.6	0.6	0.6	0.6		0.7	0.9	
Contribution from real GDP growth		-4.2	-4.8	-3.5			-3.0	-1.1	-1.2	-1.3	-1.4	-1.4		-1.5	-1.5	
Contribution from price and exchange rate changes		-14.9	-9.2	-9.0			…	…	…	…	…	…		…	…	
Residual (3-4) 3/		**21.3**	**-11.3**	**-6.6**			**-39.1**	**-1.9**	**-1.3**	**-2.1**	**-1.4**	**-1.1**		**-0.6**	**-1.0**	
o/w exceptional financing		-3.8	-4.2	-2.7			-0.4	0.0	0.0	0.0	0.0	0.0		1.5	1.8	
NPV of external debt 4/		...	...	20.7			19.8	20.4	20.3	20.3	20.6	20.8		23.3	29.1	
In percent of exports		...	...	57.3			50.5	56.6	62.6	68.8	76.1	83.4		110.2	147.1	
NPV of PPG external debt		**...**	**...**	**20.7**			**19.8**	**20.4**	**20.3**	**20.3**	**20.6**	**20.8**		**23.3**	**29.1**	
In percent of exports		...	...	57.3			50.5	56.6	62.6	68.8	76.1	83.4		110.2	147.1	
Debt service-to-exports ratio (in percent)		14.5	5.5	4.5			11.7	4.3	4.4	4.1	4.3	6.3		6.1	11.1	
PPG debt service-to-exports ratio (in percent)		14.5	5.5	4.5			11.7	4.3	4.4	4.1	4.3	6.3		6.1	11.1	
Total gross financing need (billions of U.S. dollars)		0.1	0.0	0.7			1.1	0.8	0.8	1.0	1.1	1.2		1.8	4.5	
Non-interest current account deficit that stabilizes debt ratio		-2.7	28.1	19.8			42.8	3.3	3.8	5.0	4.8	5.0		3.7	3.2	
Key macroeconomic assumptions																
Real GDP growth (in percent)		5.2	5.6	5.9	4.7	0.7	6.0	6.0	6.3	6.4	6.5	6.5	6.3	6.0	5.0	5.5
GDP deflator in US dollar terms (change in percent)		17.6	10.2	14.1	1.8	15.6	13.7	7.4	6.4	6.7	4.7	3.2	7.0	3.7	2.5	3.8
Effective interest rate (percent) 5/		1.9	1.1	1.2	2.0	0.6	1.1	2.9	3.0	2.9	2.9	2.7	2.6	2.7	3.0	2.8
Growth of exports of G&S (US dollar terms, in percent)		18.7	12.4	10.9	10.0	13.1	31.0	4.4	2.0	3.0	2.5	1.4	7.4	6.9	7.0	7.7
Growth of imports of G&S (US dollar terms, in percent)		19.1	33.2	23.4	13.1	16.6	25.5	5.1	3.7	5.7	4.1	1.8	7.6	6.2	6.7	6.6
Grant element of new public sector borrowing (in percent)		...	...	...	...	...	47.7	38.5	36.5	36.3	34.7	32.9	37.8	26.0	20.8	24.7
Memorandym item:																
Nominal GDP (billions of US dollars)		7.6	8.9	10.7			12.9	14.7	16.6	18.9	21.1	23.1		38.3	82.7	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - ρ(1+g)]/(1+g+ρ+gρ) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and ρ = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 2. Ghana: Sensitivity Analyses for Key Indicators of External Debt in the Baseline Scenario, 2006-25
(In percent)

	Projections							
	2006	2007	2008	2009	2010	2011	2016	2025
NPV of debt-to-GDP ratio								
Baseline	20	20	20	20	21	21	23	29
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	20	21	21	22	22	23	28	43
A2. New public sector loans on less favorable terms in 2007-26 2/	20	21	22	23	24	25	30	40
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	20	21	21	21	21	22	24	30
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	20	22	25	25	25	25	26	30
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	20	25	31	31	32	32	36	45
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	20	29	36	35	34	34	32	32
B5. Combination of B1-B4 using one-half standard deviation shocks	20	33	46	45	44	43	41	42
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	20	28	28	28	29	29	32	40
NPV of debt-to-exports ratio								
Baseline	51	57	63	69	76	83	110	147
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	51	58	66	74	82	91	132	219
A2. New public sector loans on less favorable terms in 2007-26 2/	51	60	69	78	88	100	143	205
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	51	57	63	69	76	83	110	147
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	51	67	88	95	103	112	139	172
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	51	57	63	69	76	83	110	147
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	51	80	111	118	126	135	150	162
B5. Combination of B1-B4 using one-half standard deviation shocks	51	79	106	113	121	129	145	160
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	51	57	63	69	76	83	110	147
Debt service ratio								
Baseline	12	4	4	4	4	6	6	11
A. Alternative Scenarios								
A1. Key variables at their historical averages in 2007-26 1/	12	5	5	5	5	8	8	17
A2. New public sector loans on less favorable terms in 2007-26 2/	12	4	4	4	5	5	7	15
B. Bound Tests								
B1. Real GDP growth at historical average minus one standard deviation in 2007-08	12	4	4	4	4	6	6	11
B2. Export value growth at historical average minus one standard deviation in 2007-08 3/	12	5	5	5	5	8	8	13
B3. US dollar GDP deflator at historical average minus one standard deviation in 2007-08	12	4	4	4	4	6	6	11
B4. Net non-debt creating flows at historical average minus one standard deviation in 2007-08 4/	12	4	5	6	6	8	10	13
B5. Combination of B1-B4 using one-half standard deviation shocks	12	4	5	6	6	8	9	13
B6. One-time 30 percent nominal depreciation relative to the baseline in 2007 5/	12	4	4	4	4	6	6	11
Memorandum item:								
Grant element assumed on residual financing (i.e., financing required above baseline) 6/	23	23	23	23	23	23	23	23

Source: Staff projections and simulations.

1/ Variables include real GDP growth, growth of GDP deflator (in U.S. dollar terms), non-interest current account in percent of GDP, and non-debt creating flows.
2/ Assumes that the interest rate on new borrowing is by 2 percentage points higher than in the baseline., while grace and maturity periods are the same as in the baseline.
3/ Exports values are assumed to remain permanently at the lower level, but the current account as a share of GDP is assumed to return to its baseline level after the shock (implicitly assuming an offsetting adjustment in import levels).
4/ Includes official and private transfers and FDI.
5/ Depreciation is defined as percentage decline in dollar/local currency rate, such that it never exceeds 100 percent.
6/ Applies to all stress scenarios except for A2 (less favorable financing) in which the terms on all new financing are as specified in footnote 2.

Table 3. Ghana: External Debt Sustainability Framework, Alternative Scenario A, 2003-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical	Standard	Projections									
	2003	2004	2005	Average 6/	Deviation 6/	2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2025	2012-25 Average
External debt (nominal) 1/	**99.0**	**72.7**	**59.2**			**22.3**	**23.6**	**27.3**	**31.1**	**34.1**	**33.6**		**31.5**	**33.7**	
o/w public and publicly guaranteed (PPG)	99.0	72.7	59.2			22.3	23.6	27.3	31.1	34.1	33.6		31.5	33.7	
Change in external debt	-0.5	-26.3	-13.5			-36.9	1.2	3.7	3.8	3.0	-0.5		-0.4	0.4	
Identified net debt-creating flows	-17.9	-15.0	-6.8			0.1	3.0	2.4	1.6	0.6	-0.3		0.5	0.6	
Non-interest current account deficit	**0.7**	**1.7**	**7.0**	**5.0**	**4.1**	**4.4**	**4.3**	**4.6**	**4.6**	**4.2**	**3.7**		**3.9**	**3.1**	3.7
Deficit in balance of goods and services	15.9	21.1	25.6			24.6	23.4	21.7	19.8	18.2	16.4		12.9	8.4	
Exports	40.7	39.3	36.1			39.2	37.4	35.1	32.9	31.4	30.4		24.2	21.3	
Imports	56.6	60.4	61.7			63.8	60.8	56.8	52.7	49.6	46.7		37.2	29.7	
Net current transfers (negative = inflow)	-15.7	-20.6	-19.7	-13.3	4.9	-20.1	-19.2	-17.3	-15.3	-14.0	-12.7		-9.1	-5.2	-8.0
Other current account flows (negative = net inflow)	0.4	1.3	1.2			0.0	0.1	0.1	0.1	0.0	0.1		0.1	-0.1	
Net FDI (negative = inflow)	**-1.0**	**-3.7**	**-1.4**	**-1.5**	**0.9**	**-2.1**	**-0.7**	**-1.4**	**-2.0**	**-2.5**	**-2.9**		**-2.5**	**-2.2**	-2.4
Endogenous debt dynamics 2/	**-17.6**	**-13.0**	**-12.5**			**-2.3**	**-0.6**	**-0.8**	**-0.9**	**-1.1**	**-1.1**		**-0.9**	**-0.3**	
Contribution from nominal interest rate	1.5	0.9	0.0			0.6	0.6	0.7	1.0	1.3	1.5		1.2	1.2	
Contribution from real GDP growth	-4.2	-4.8	-3.5			-3.0	-1.2	-1.5	-1.9	-2.3	-2.6		-2.1	-1.6	
Contribution from price and exchange rate changes	-14.9	-9.2	-9.0			...	...	...	...	...	...		...	...	
Residual (3-4) 3/	**17.4**	**-11.3**	**-6.6**			**-36.9**	**-1.8**	**1.4**	**2.2**	**2.4**	**-0.2**		**-0.8**	**-0.2**	
o/w exceptional financing	-3.8	-6.8	-1.6			-0.4	0.0	0.0	0.0	0.0	0.0		0.0	0.0	
NPV of external debt 4/	...	...	21.3			19.9	20.5	24.2	28.3	31.3	30.2		27.4	30.3	
In percent of exports	...	...	59.0			50.8	54.7	69.0	86.0	99.9	99.6		112.9	142.3	
NPV of PPG external debt	**...**	**...**	**21.3**			**19.9**	**20.5**	**24.2**	**28.3**	**31.3**	**30.2**		**27.4**	**30.3**	
In percent of exports	...	...	59.0			50.8	54.7	69.0	86.0	99.9	99.6		112.9	142.3	
Debt service-to-exports ratio (in percent)	14.5	5.5	2.4			12.0	4.4	4.4	4.7	5.7	7.5		11.7	13.9	
PPG debt service-to-exports ratio (in percent)	14.5	5.5	2.4			12.0	4.4	4.4	4.7	5.7	7.5		11.7	13.9	
Total gross financing need (billions of U.S. dollars)	0.4	0.0	0.7			0.9	0.9	0.9	1.0	1.0	1.0		2.2	4.7	
Non-interest current account deficit that stabilizes debt ratio	1.2	28.1	20.5			41.3	3.1	0.8	0.8	1.2	4.2		4.3	2.7	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.2	5.6	5.9	4.7	0.7	6.0	6.2	7.0	8.0	8.5	8.5	7.4	7.3	5.0	6.3
GDP deflator in US dollar terms (change in percent)	17.6	10.2	14.1	1.8	15.6	13.7	7.4	6.4	6.7	4.7	3.2	7.0	3.7	2.5	3.8
Effective interest rate (percent) 5/	1.9	1.1	0.0	1.9	0.8	1.3	3.2	3.4	4.0	4.6	4.9	3.6	4.2	4.0	4.1
Growth of exports of G&S (US dollar terms, in percent)	18.7	12.4	10.9	10.0	13.1	31.0	8.8	6.7	8.1	8.4	8.4	11.9	7.8	7.6	7.6
Growth of imports of G&S (US dollar terms, in percent)	19.9	24.1	23.4	12.9	15.1	24.7	8.7	6.4	7.0	6.8	5.6	9.9	7.2	6.3	6.9
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	28.3	32.4	10.5	5.6	6.4	27.2	18.4	15.0	10.9	14.1
Memorandym item:															
Nominal GDP (billions of US dollars) ...	7.6	8.9	10.7			12.9	14.7	16.8	19.3	22.0	24.6		43.8	98.0	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
4/ Assumes that NPV of private sector debt is equivalent to its face value.
5/ Current-year interest payments devided by previous period debt stock.
6/ Historical averages and standard deviations are generally derived over the past 10 years, subject to data availability.

Table 4. Ghana: External Debt Sustainability Framework, Alternative Scenario B, 2003-2025 1/
(In percent of GDP, unless otherwise indicated)

	Actual			Historical Average 6/	Standard Deviation 6/	Projections									
	2003	2004	2005			2006	2007	2008	2009	2010	2011	2006-11 Average	2016	2025	2012-25 Average
External debt (nominal) 1/	**99.0**	**72.7**	**59.2**			**22.3**	**23.6**	**27.5**	**31.8**	**35.5**	**35.7**		**36.0**	**40.0**	
o/w public and publicly guaranteed (PPG)	99.0	72.7	59.2			22.3	23.6	27.5	31.8	35.5	35.7		36.0	40.0	
Change in external debt	-0.5	-26.3	-13.5			-36.9	1.3	3.9	4.3	3.7	0.2		0.0	0.5	
Identified net debt-creating flows	-21.8	-15.0	-6.8			0.1	4.2	4.1	3.9	3.0	1.8		1.7	-1.0	
Non-interest current account deficit	**-3.2**	**1.7**	**7.0**	**4.2**	**5.2**	**4.4**	**5.5**	**6.7**	**6.9**	**6.4**	**5.4**		**5.7**	**1.4**	4.3
Deficit in balance of goods and services	12.0	21.1	25.6			24.8	24.6	24.0	22.5	21.0	18.6		15.8	8.6	
Exports	40.7	39.3	36.1			39.2	36.3	33.2	30.1	27.9	26.2		20.7	16.3	
Imports	52.7	60.4	61.7			64.1	60.9	57.2	52.7	48.9	44.7		36.5	24.9	
Net current transfers (negative = inflow)	-15.7	-20.6	-19.7	-13.3	4.9	-20.1	-19.3	-17.4	-15.7	-14.6	-13.5		-10.6	-7.6	-9.6
Other current account flows (negative = net inflow)	0.4	1.3	1.2			-0.3	0.2	0.1	0.1	0.1	0.4		0.5	0.5	
Net FDI (negative = inflow)	**-1.0**	**-3.7**	**-1.4**	**-1.5**	**0.9**	**-2.1**	**-0.7**	**-2.0**	**-2.4**	**-2.8**	**-3.1**		**-3.4**	**-2.0**	-2.9
Endogenous debt dynamics 2/	**-17.6**	**-13.0**	**-12.5**			**-2.3**	**-0.5**	**-0.6**	**-0.6**	**-0.6**	**-0.5**		**-0.6**	**-0.4**	
Contribution from nominal interest rate	1.5	0.9	0.0			0.6	0.6	0.7	1.0	1.3	1.6		1.4	1.5	
Contribution from real GDP growth	-4.2	-4.8	-3.5			-3.0	-1.2	-1.3	-1.6	-1.9	-2.1		-2.0	-1.8	
Contribution from price and exchange rate changes	-14.9	-9.2	-9.0			…	…	…	…	…	…		…	…	
Residual (3-4) 3/	**21.3**	**-11.3**	**-6.6**			**-36.9**	**-2.9**	**-0.2**	**0.4**	**0.6**	**-1.6**		**-1.7**	**1.5**	
o/w exceptional financing	-3.8	-4.2	-2.7			-1.7	0.0	0.0	0.0	0.0	0.0		1.5	1.8	
NPV of external debt 4/	...	...	21.3			19.9	20.5	24.4	29.0	32.7	32.2		31.3	35.9	
In percent of exports	...	...	59.0			50.8	56.5	73.6	96.2	117.1	122.9		150.9	219.9	
NPV of PPG external debt	**...**	**...**	**21.3**			**19.9**	**20.5**	**24.4**	**29.0**	**32.7**	**32.2**		**31.3**	**35.9**	
In percent of exports	...	...	59.0			50.8	56.5	73.6	96.2	117.1	122.9		150.9	219.9	
Debt service-to-exports ratio (in percent)	14.5	5.5	2.4			12.0	4.5	4.7	5.2	6.7	9.2		15.7	21.5	
PPG debt service-to-exports ratio (in percent)	14.5	5.5	2.4			12.0	4.5	4.7	5.2	6.7	9.2		15.7	21.5	
Total gross financing need (billions of U.S. dollars)	0.1	0.0	0.7			0.9	1.0	1.2	1.3	1.3	1.3		2.5	3.3	
Non-interest current account deficit that stabilizes debt ratio	-2.7	28.1	20.5			41.3	4.2	2.8	2.6	2.7	5.2		5.7	0.9	
Key macroeconomic assumptions															
Real GDP growth (in percent)	5.2	5.6	5.9	4.7	0.7	6.0	6.0	6.3	6.4	6.5	6.5	6.3	6.0	5.0	5.5
GDP deflator in US dollar terms (change in percent)	17.6	10.2	14.1	1.8	15.6	13.7	7.4	6.4	6.7	4.7	3.2	7.0	3.7	2.5	3.8
Effective interest rate (percent) 5/	1.9	1.1	0.0	1.9	0.8	1.3	3.2	3.4	4.0	4.6	4.9	3.6	4.2	4.0	4.1
Growth of exports of G&S (US dollar terms, in percent)	18.7	12.4	10.9	10.0	13.1	31.0	5.4	3.3	3.0	3.4	3.0	8.2	7.7	6.4	5.9
Growth of imports of G&S (US dollar terms, in percent)	19.1	33.2	23.4	13.1	16.6	25.2	8.2	6.2	4.5	3.5	0.6	8.0	5.2	3.7	4.8
Grant element of new public sector borrowing (in percent)	...	...	...	...	...	28.3	32.4	10.5	5.6	6.4	27.2	18.4	15.0	10.9	14.1
Memorandym item:															
Nominal GDP (billions of US dollars)	7.6	8.9	10.7			12.9	14.7	16.6	18.9	21.1	23.1		38.3	82.7	

Source: Staff simulations.

1/ Includes both public and private sector external debt.
2/ Derived as [r - g - r(1+g)]/(1+g+r+gr) times previous period debt ratio, with r = nominal interest rate; g = real GDP growth rate, and r = growth rate of GDP deflator in U.S. dollar terms.
3/ Includes exceptional financing (i.e., changes in arrears and debt relief); changes in gross foreign assets; and valuation adjustments. For projections also includes contribution from price and exchange rate changes.
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